# Remote Sensing Image Ship Detection Based on Dynamic Adjusting Labels Strategy

Chaofan Pan, Runsheng Li, Wei Liu, Wanjie Lu, Chaoyang Niu, and Quanfu Bao

***Abstract*— Remote sensing ship detection is a hotspot in computer vision, which is vital in both military and civilian fields. Nevertheless, the arbitrary orientation and dense arrangement of ship targets impose significant challenges in high-precision detection. Although research on this problem has progressed, high-precision detection is still limited by angular prediction accuracy. To tackle this issue, we start with angle prediction and propose a dynamic adjusting labels (DAL) strategy based on binary coded label (BCL). DAL strategy dynamically adjusts the ground-truth coded labels in the training process to guide angle coding for tendency learning. This strengthens the coupling between the angle coding bits and improves the performance of small granularity intervals. Due to the angle interval granularity difference, the learning difficulty of the coding layers and the convergence speed vary significantly. Aiming at this problem, we add a gradient truncation mechanism to each coding bit loss. The mechanism can effectively balance the coding layers' learning strength and enhance the model's training emphasis on coding bits corresponding to small granularity intervals, thus avoiding the effect of coding bits learning imbalance on angle prediction. Extensive experiments based on three public datasets demonstrate our method's superiority in high-precision detection and the state-of-the-art performance. Our code is available at https://github.com/lirunsheng2008/SDALS.**

***Index Terms*— Angle coding bits, arbitrary orientation, dense arrangement, gradient truncation mechanism, remote sensing images, ship detection.**

## I. Introduction

REMOTE sensing is critical for Earth reconnaissance, with ships being important remote sensing targets. Indeed, accurate ship detection plays a vital role in economic development, national defense security, and overseas interests. Ship detection utilizing remote sensing images includes synthetic aperture radar (SAR) and optical images. SAR can realize all-day and all-weather observation, covering an extensive range, so ship detection based on SAR images is widely adopted, with many scholars achieving fruitful results [1], [2], [3], [4], [5]. However, the low resolution of SAR images and the lack of details such as texture and color significantly challenge the accurate positioning and classification of ship targets. Although some scholars have studied the high-precision location of ships in SAR images [6], [7], [8], [9], [10], [11], [12], the higher precise classification still faces some difficulties. In contrast, optical images contain rich details and high resolution, which is conducive to more accurate target detection tasks. Early ship detection methods mainly relied on handcrafted features [13], [14], [15], [16], [17], [18], [19], [20], [21]. These methods extracted regions of interest (RoIs) and identified targets by manually modeling features, but their feature characterization ability and low efficiency were limited. Nevertheless, the rapid development of deep learning technology has significantly improved the object detection, suggesting more efficient detection algorithms.



(a) (b)

Fig. 1. Comparison of the: (a) horizontal and (b) rotated bounding boxes.

Unlike traditional methods, deep learning detection algorithms based on convolutional neural networks (CNNs) [22] can obtain more discriminant features by extracting deep semantic information, thus significantly improving the detection accuracy. Region proposals with convolution neural networks (RCNNs) [23] and Fast R-CNN [24] are the pioneering work of deep learning object detection algorithms. These methods exploited the CNN's feature extraction capability to simplify the detection and improve the algorithm's accuracy and efficiency. On this basis, object detection based on deep learning has gradually developed into a mature system involving feature extraction, feature enhancement, position regression, and classification recognition [25], [26], [27], [28], [29], [30], [31], [32], [33], [34], [35], [36].

The original object detection algorithm types, i.e., RCNN [23], [24], [25], Yolo [30], [31], [32], and single shot multibox detector (SSD) [34], [35], [37], use a horizontal bounding box to locate the target by regressing the $(x, y, w, h)$ parameters that define the target's center coordinates and size. Although this strategy achieves appealing results in object detection within natural scenes, remote sensing images are captured from an overhead perspective, with targets presenting

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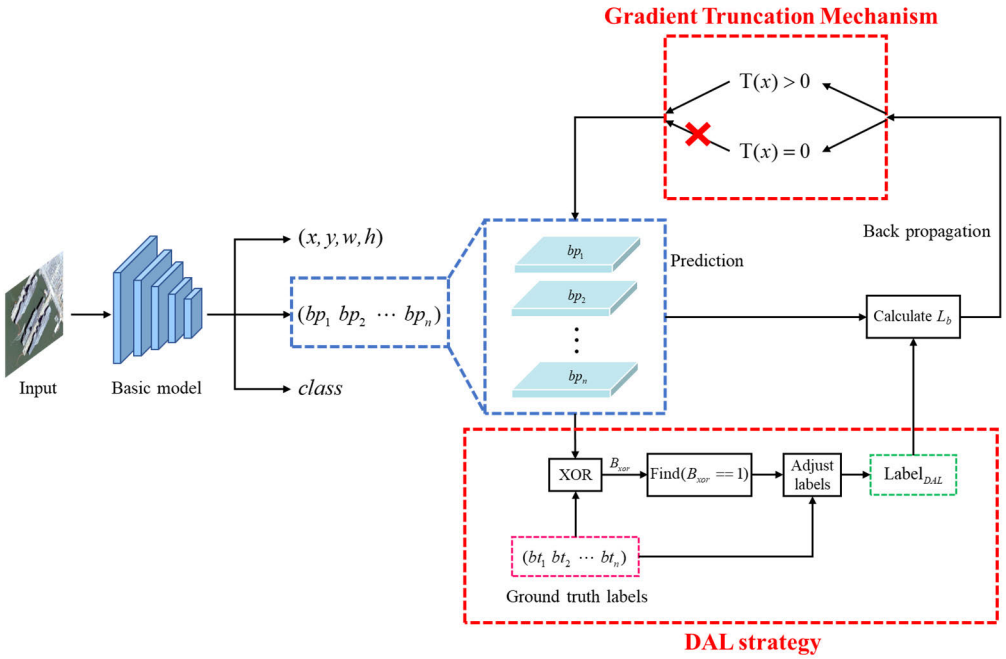



Fig. 2. Proposed detection framework. Both DAL and gradient truncation mechanism act on the angle coding layers of the basic model. DAL dynamically adjusts the coding labels to the optimal guidance direction according to the angle coding predicted value of the training process. Gradient truncation mechanism balances coding layer's learning intensity by controlling the gradient's backpropagation to prevent the overfitting of simple coding bits.

arbitrary orientations. In this case, the horizontal bounding box cannot effectively fit the targets, especially those with large aspect ratios such as ships, introducing many background areas and significantly affecting the detection accuracy [Fig. 1(a)]. Unlike the horizontal bounding box, the rotated bounding box considers the targets' directional information, affording accurate positioning and thus reducing the background factors' influence [Fig. 1(b)]. Therefore, the rotation-based object detection has attracted significant research attention.

The most direct modeling method for rotated bounding boxes involves adding angle parameters based on $(x, y, w, h)$ to form a five-parameter system $(x, y, w, h, \theta)$, where $x$, $y$, $w$, $h$, and $\theta$ indicate the box's center point coordinates, size, and angle. In order to achieve the regression prediction of $(x, y, w, h, \theta)$, Ma et al. [38] proposed the rotation region proposal network (RRPN) that exploited a region proposal network (RPN) [25] combined with a rotation region-of-interest (RRoI) pooling [39] scheme to collect the features in a rotated RoI proposed by RRPN. Although RRPN can better adapt to multioriented scene objects than horizontal-anchor-based RPN, the rotated anchors' angle sequence that is manually set is discrete and thus cannot guarantee the matching between the rotated anchors and the arbitrary-oriented ground-truth boxes. Hence, this solution is not conducive to the angle regression. Additionally, the strategy of angle-intensive sampling will consume significant computing resources and slow down the network speed. Therefore, Ding et al. [40] solved the contradiction between angle sampling and network speed in RRPN by proposing the RoI transformer. This method utilizes RPN as the starting point and realizes feature extraction by applying spatial transformation on the horizontal RoI, as generated by RPN. This strategy achieves feature alignment between the RoI and target region and avoids setting up many rotated anchors that slow down the network. Yang et al. [41] developed a two-staged rotation region detection network (RDN) based on RPN and Fast-RCNN. RDN exploits the rotated anchors to generate rotated regions and then extracts the features of their horizontal outer regions to obtain fixed-length feature vectors. Finally, the five-parameter prediction results $(x, y, w, h, \theta)$ are obtained through the fully connected layers.

A typical problem that angle regression faces are the boundary discontinuity caused by angle periodicity, with current solutions focusing on the loss function. For example, Yang et al. [42] added a constant factor $(|-\log(\text{IoU})|/|\text{Smooth}L_1|)$ for the Smooth$L1$ loss and used boundary case properties $|-\log(\text{IoU})| \approx 0$ to avoid the boundary problem, where IoU is the intersection over union between two boxes. The modulated rotation loss [43] achieves loss continuity at the boundary by adding a correction term that performs mirror symmetry treatment on the angle loss at the boundary position. The pixels-IoU (PIoU) loss [44] calculates the IoU of the rotated bounding boxes by counting pixels through a smoothing kernel function and performs an IoU optimization process from a global perspective.

Nevertheless, the above methods are complex and remedial, unable to eliminate the boundary discontinuity. Therefore, Yang and Yan [45] developed an angle prediction method based on classification that divides the angle definition domain into multiple intervals, which are encoded. To realize the angle distance perception, the authors designed the circular smooth label (CSL) to smooth the interval coding and measure the angle distance, eliminating the boundary problem of angle prediction. CSL is a sparse coding method that performs one-hot coding prediction by dividing the angle range from 0 to 180 into multiple intervals, and each bit of coding is predicted

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through a coding layer. In order to ensure the accuracy, the thickness of the coding layer is large. For example, if the angle range is equally divided into 180 intervals, the thickness of the coding layer is 180, the length of each interval is 1, and the maximum angle error is 0.5. However, this CSL-based technique requires heavy coding layers to guarantee the accuracy, thus significantly affecting the calculation efficiency. Subsequently, Yang et al. [46] proposed densely coded label (DCL) with binary coded label (BCL) [47] and gray coded label (GCL) [48], which reduced the coding layers' thickness and improved the model's execution speed. It should be noted that dense coding is equivalent to converting decimal angles into binary, e.g., the 4-bit coding 0000-1111 can represent the 0–15 range. Hence, using 16-bit coding for prediction is no longer required, thus reducing the number of coding layers. DCL relies on base conversion relation for efficient coding and designs angle distance and aspect ratio sensitive weighting (ADARSW) to train the coding layers, thus achieving appealing results.

This article aims to find a rotating detector suitable for ship targets without an angle boundary discontinuity. By sorting out the current methods presented above, we argue that rotation-based detection using rotated bounding boxes is a reasonable choice for objects with a large aspect ratio, such as ships. Furthermore, the angle classification method avoids boundary problems caused by angle periodicity. Therefore, we propose a rotation detection algorithm suitable for ship targets based on the deep thinking part of BCL in DCL. The proposed method reinterprets BCL's encoding and decoding processes from a geometry perspective and then uses the dynamic adjusting labels (DAL) strategy to guide the angle coding for tendency learning. This strategy aims to solve the problem of DCL's poor coding bit coupling. In addition, the different granularity of angle intervals represented by each coding bit significantly affects the learning difficulty of the coding layers. The lower the coding bit, the smaller the granularity of the corresponding angle interval, and the more complex the learning. To solve this problem, we add a gradient truncation mechanism to the angle coding loss, thus preventing the model from over-fitting the simple coding bits and assisting the model in focusing on learning the difficult coding bits. The main contributions of this article are as follows.

1) BCL has a spatial significance. From the geometry perspective, we reveal that the corresponding coding and decoding processes involve angle search using dichotomy. Thus, we rewrite the coding and decoding algorithm based on dichotomy.
2) The correlation between the BCL coding bits is insufficient, and the training process between the coding bits is relatively independent. Hence, we propose a DAL strategy to dynamically adjust the label according to the network's coding prediction. DAL realizes the oriented guided learning and splits the point-to-point training mode to pay more attention to the overall coding coupling and prediction tendency.
3) Due to the differences in learning difficulty per coding bit, we set a gradient truncation mechanism for the angle loss, thus ensuring the overall learning effect of the network on the coding layers and preventing simple coding bits from over-fit due to neglecting to learn the difficult bits.

The remainder of this article is organized as follows. Section II reviews the relevant work, and Section III introduces the proposed method's details. Section IV provides the dataset's information, designs experiments to verify our method's effectiveness, and visualizes the detection results. Additionally, this section evaluates and discusses the algorithm's influence on the accuracy distribution and angle learning. Finally, Section V summarizes this work.

## II. Related Works

### A. Horizontal Object Detection

As one of the critical tasks in computer vision, object detection locates the target area using bounding boxes and obtains target location information. According to different classification criteria, object detectors based on CNN can be divided into two-stage and one-stage, and anchor-based and anchor-free detectors. The two-stage object detectors [23], [24], [25], [26], [27], [28], [29], [49] generate a series of candidate region proposals through RPN and then perform subsequent classification and regression prediction. These methods achieve good results, but their speed is slow. The one-stage detectors [30], [31], [32], [33], [34], [35], [36], [50], [51], [52] improve the speed by directly predicting the object class probability and coordinates at the cost of sacrificing accuracy. Anchor-based detectors match positive and negative samples using anchors and adopt a regression process by predicting the offsets relative to anchors, thus narrowing the regression range. However, these methods suffer from substantial computational complexity and low efficiency. In contrast, the anchor-free methods [53], [54], [55] abandon the anchor concept and express the target through anchor points or key points, which is simpler and more efficient. Nevertheless, point prediction has ample solution space, thus leading to a severe imbalance of positive and negative samples.

With the development of object detection technology, the boundary between the above methods has gradually blurred. Some fusion models [56], [57], [58] have been proposed to solve the shortcomings of the previous methods and improve the detection performance. Still, these methods locate targets through a horizontal bounding box, which has limited adaptability to arbitrary-oriented object detection.

### B. Arbitrary-Oriented Object Detection

Oriented object detectors have strong applicability for arbitrary object detection in remote sensing scenarios and text in natural scenes. For some densely arranged targets with a large aspect ratio, the horizontal bounding box will introduce many background noise areas. In contrast, the rotating bounding box considers the target direction information and thus effectively reduces the influence of complex background. Some early methods [38], [59] expanded horizontal detection by setting rotating anchors to sample the target angle. However, this discrete sequence of angles does not guarantee a high-quality match for any arbitrary distribution of angles,

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posing a nonconducive solution for the angle regression prediction, introducing many redundant boxes, and forcing an imbalance between positive and negative samples. The RoI transformer [40] uses an RRoI leaner module to learn the transformation from horizontal RoI to rotary RoI. Then, it extracts rotary invariant features from the rotary RoI through RRoI wrapping that is used in the subsequent classification and regression, solving the contradiction between angle sampling and calculating the rotated prior anchors. This rotation detection idea through spatial transformation can be traced back to the text detection model rotational region CNN (R2CNN) [60]. This design can not only effectively alleviate the misalignment between the RoI and the target, but also avoid the low detection efficiency caused by many designs of rotated anchors. The RoI transformer is a three-stage algorithm with significant advantages in detection accuracy, but it is slow and thus is applied in situations where high detection speed is not required. In addition, the one-stage detector refined single-stage detector (R3det) achieves accuracy and speed using a progressive regression approach from coarse to fine granularity.

Some methods [41], [42], [61] consider the feature level and exploit the spatial misalignment between the convolution feature and the target position as an important reason affecting the positioning accuracy. These methods then extract alignment features. Other works [62], [63], [64] recognize that there are attribute contradictions in the features required for classification and regression tasks, so the rotation-sensitive features that are more suitable for localization are extracted by task decoupling.

Because of the angle's periodicity, boundary problems exist in the angle's regression prediction. In this regard, Yang and Yan [45] and Yang et al. [46] proposed a new idea based on angle classification and solved the above problems through coding. Sun et al. [65] combined angle classification and bidirectional feature fusion to improve ship target detection accuracy. Inspired by this, we think more deeply and propose a new method to guide the angle coding layers learning, aiming to improve the model's detection performance in the high-precision intervals.

## III. Proposed Method

This section discusses our method using BCL with geometric significance. Specifically, first, we reinterpret BCL from the perspective of geometry, show the essence of angle prediction through an intuitive coding form, and then deduce the encoding and decoding processes. Then, aiming at feature information isolation and the angle learning difficulties caused by coding the bit's independence, we propose a DAL strategy to increase the intercode coupling. Finally, we perform gradient truncation on the loss function to prevent the network from overfitting the simple samples and focus more on learning the difficult samples. Fig. 2 illustrates our method's pipeline.

### A. Geometric Meaning of BCL

BCL is the binary representation of decimal angles. Each coding bit of BCL contains two states representing the angle's spatial region. Considering the angle bisector as the boundary,

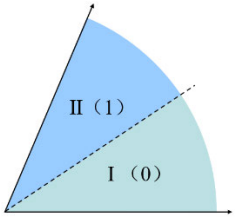



Fig. 3. Geometric meaning of a coding bit. Coding bit "0" means the angle is in subinterval I, while Coding bit "1" means the angle is in subinterval II.

"0" represents the right subinterval of the angle bisector, and "1" is the left subinterval (Fig. 3).

**Algorithm 1** Binary Coded Label (BCL)

**Input:** code length $n$, angle $A$.
**Output:** a list $L$ containing a binary coded label's bits.

```
L = []
# the middle position after each binary search.
Bisector = 90
# Coding procedure
for i in range(n) do
    # Determine the magnitude relationship between A
    and Bisector.
    b = Sign(A - Bisector)
    # Update the middle position.
    Bisector = Bisector + 90 * 0.5^i * b
    # Converts '−1' to '0' in the output value of the sign
    function.
    if b < 0 then
        b = b + 1
    end
    L.append(b)
end
Return L
```

Therefore, BCL can be understood as an angle search process using a binary search method, and the value of the coding bits represents the search direction (Fig. 4). The size of the angle region is halved with each search operation. Therefore, the granularity $\omega$ of the angle interval decreases exponentially with the increase of the code length $n$. Suppose the angle range is $0 \sim 180°$, the relationship between $\omega$ and $n$ is $\omega = (180/2^n)$. Each search is bounded by the angle bisector of the angle region. The encoding process is presented in Algorithm 16, and the decoding process is defined as follows:

$$\begin{aligned} A &= 90 \cdot \left(1 - \sum_{i=1}^{n} 0.5^i \cdot (-1)^{b_i}\right) \\ &= 90 \cdot \left(1 + \sum_{i=1}^{n} 0.5^i \cdot (2b_i - 1)\right) \\ &= 90 \cdot \left(1 - \frac{0.5 \cdot (1 - 0.5^n)}{1\text{-}0.5} + \sum_{i=1}^{n} 0.5^i \cdot 2b_i\right) \\ &= 90 \cdot \left(0.5^n + \sum_{i=1}^{n} 0.5^i \cdot 2b_i\right) \end{aligned} \tag{1}$$

where $b_i$ denotes the $i$th coding bit and $i = 1, 2, \ldots, n$.

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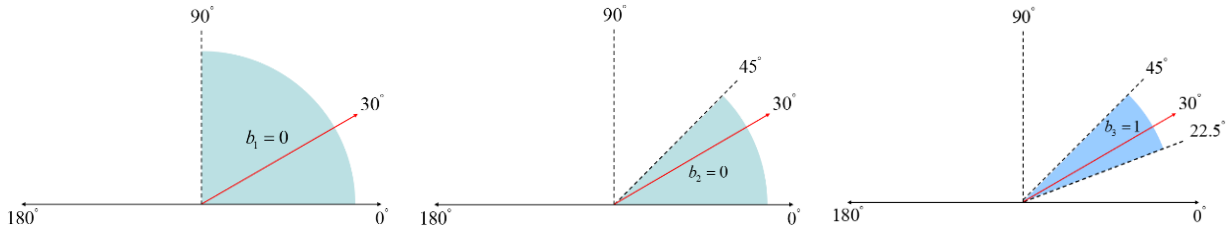



Fig. 4. Angle coding utilizing dichotomy search. The search process is to segment the angle interval through the angle bisector and assign the value of "0" or "1" to each coding bit according to the subinterval where the angle is located.

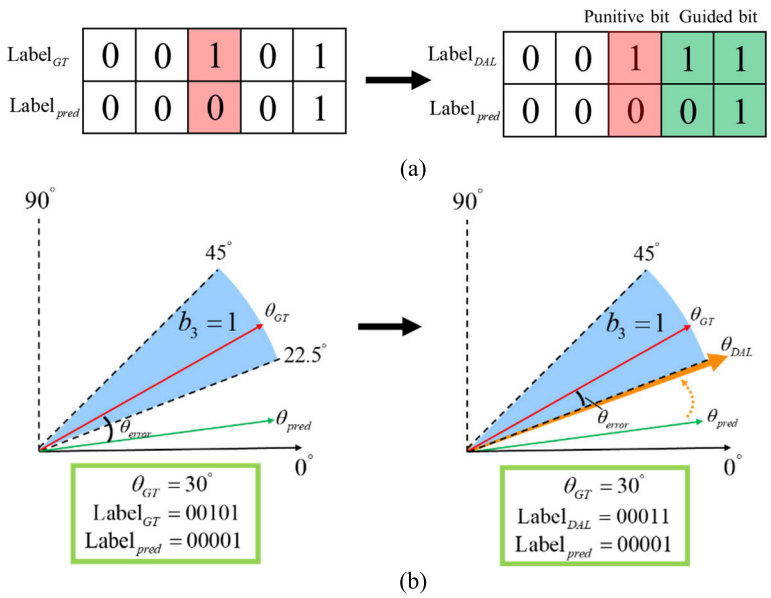



Fig. 5. DAL principle: (a) coded label adjustment in the training process and (b) geometric operations.

### B. DAL Strategy

In BCL, the angle interval granularity size decreases exponentially with the coding bit from high to low. The smaller the interval, the more difficult the coding layer is to learn.

Therefore, ensuring the effective learning of each coding layer is challenging. We argue that the independence of the angle coding layers imposes insufficient information interaction and low effective angle perception. In this case, a thick coding layer is counterproductive, confirming Yang et al. [46].

To enhance the coupling between the coding bits, we design a DAL strategy, defined as

$$\begin{cases} B_{\mathrm{XOR}} = (\mathrm{bt}_1 \mathrm{bt}_2 \cdots \mathrm{bt}_n) \oplus (\mathrm{bp}_1 \mathrm{bp}_2 \cdots \mathrm{bp}_n) \\ \mathrm{index} = \mathrm{Find}(B_{\mathrm{XOR}} == 1) \\ i \in \{\mathrm{index}[0]+1, \ldots, n\} \\ \mathrm{bt}_i = \begin{cases} \mathrm{bt}_{\mathrm{index}[0]}, & \text{otherwise} \\ 1 - \mathrm{bt}_{\mathrm{index}[0]}, & \mathrm{index}[0] == 1 \quad \text{and} \quad \mathrm{bt}_1 == \mathrm{bt}_2 \end{cases} \end{cases} \tag{2}$$

where $\mathrm{bt}_i$ and $\mathrm{bp}_i$ indicate the ground truth and the predicted coded labels, respectively, and $B_{\mathrm{XOR}}$ is the XOR operation result of $\mathrm{bt}_i$ and $\mathrm{bp}_i$.

The basic idea of DAL is to optimally guide the angle coding layers' training process by adjusting the ground-truth coding labels based on network prediction. Traditional point-to-point learning is transformed into tendency learning, breaking the stereotype of training strictly following the ground-truth label. This new model pays more attention to the overall coding connection and the global nature of learning, minimizing the prediction error. For completeness, tendency learning refers to guiding subsequent codes to learn in the direction of minimum error when a specific code bit is incorrectly predicted rather than focusing only on the consistency of each code bit's predicted value and true value.

Specifically, assuming that the coded label length $n$ is 5, the ground-truth coded labels are $\mathrm{bt}_1$ $\mathrm{bt}_2$ $\mathrm{bt}_3$ $\mathrm{bt}_4$ $\mathrm{bt}_5$, the predicted coded labels are $\mathrm{bp}_1$ $\mathrm{bp}_2$ $\mathrm{bp}_3$ $\mathrm{bp}_4$ $\mathrm{bp}_5$, and the first error coded position is $b_3$. Then, the ground-truth coding bit $\mathrm{bt}_3$ is adjusted to $\mathrm{bp}_3$, and all ground-truth coded labels after $b_3$ are adjusted to $\mathrm{bt}_3$, ultimately providing $\mathrm{bt}_1$ $\mathrm{bt}_2$ $\mathrm{bp}_3$ $\mathrm{bt}_3$ $\mathrm{bt}_3$. During training, $b_4$ and $b_5$ are used as guidance bits, while $b_3$ is a punishment bit for calculating the loss against the original ground-truth label and preventing the positive incentive effect on the wrong coding bits. Therefore, the ground-truth coded label at $b_3$ can still be regarded as $\mathrm{bt}_3$ when calculating the loss, as shown in Fig. 5(a). This method aims to correlate the coding bits and guide them for tendentious learning. When a specific coding bit is incorrectly predicted, the subsequent bits are dynamically adjusted according to the minimization principle of the angle error, as shown in Fig. 5(b), thus

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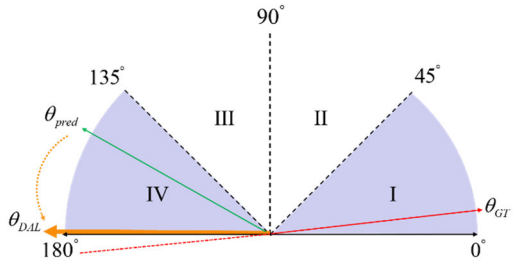



Fig. 6. Optimal guidance direction. The optimal guiding direction is the horizontal direction when the ground truth is located in subinterval I or IV and the prediction of a bit at $b_1$ is wrong.

avoiding the problems of feature information isolation and low learning efficiency caused by truth-only training and coding bits independence. To enhance the explanation, it should be noted that the schematic of the minimization principle is illustrated in Fig. 5(b), highlighting that when a specific code point is wrong, the following code points are appropriately adjusted to minimize the error.

If $b_k$ is predicted by mistake and the subsequent coding bits are still predicted according to the ground-truth label, the error between prediction and ground truth is defined as

$$\text{error}_\theta = 90 \cdot 0.5^{k-1}. \tag{3}$$

After DAL's uniform directional guidance to lower coding bits than $b_k$, the prediction error and bit serial number are no longer unique, and the error values are

$$\text{error}_\theta \in \left\{ e | e = 90 \cdot 0.5^{i-1}, i = k, k+1, \ldots, n \right\} \tag{4}$$

where the maximum and minimum prediction errors are $\text{error}_{\max} = 90 \cdot 0.5^{k-1}$ and $\text{error}_{\min} = 90 \cdot 0.5^{n-1}$, respectively.

It should be noted that when $\text{bt}_1 = \text{bt}_2$, the ground-truth angle is in subinterval I or subinterval IV, as illustrated in Fig. 6. If the prediction of a bit at $b_1$ is wrong, the optimal guidance direction for the remaining coding bits is horizontal, and DAL will uniformly adjust the coded labels after $b_1$ to $1 - \text{bt}_1$ instead of $\text{bt}_1$. This is because there always exists a subinterval between the ground-truth angle and the predicted angle, and there is no common boundary in the interval $(0, 180°)$ between the subintervals where they are located. Therefore, the optimal guiding direction can only be horizontal direction.

The principle is as follows: assuming that the ground-truth angle $\theta_{\text{GT}}$ is located in subinterval I, then $\text{bp}_1 = 1$, and the guiding direction is located in the interval $90° \sim 180°$. For each guiding direction $\theta_{\text{guide}}$, its error value relative to $\theta_{\text{GT}}$ is defined as

$$\text{error}_\theta = \min\left\{ \theta_{\text{guide}} - \theta_{\text{GT}}, \theta_{\text{GT}} + 180 - \theta_{\text{guide}} \right\}. \tag{5}$$

The guidance error distribution is depicted in Fig. 7(a), where the vertical coordinate value represents the error distance between $\theta_{\text{guide}}$ and $\theta_{\text{GT}}$. The average error corresponding to $\theta_{\text{guide}}$ can be obtained by integrating the error distribution along the $\theta_{\text{GT}}$ axis and calculating the mean value, which is used to measure the prediction accuracy of the guidance direction in the true value region. Since $\theta_{\text{GT}} \in (0, 45)$, we calculate the average error of $\theta_{\text{guide}}$ as follows:

$$\Delta\theta_{\text{a}} = \frac{1}{45} \int_0^{45} \text{error}_\theta \, d\theta_{\text{GT}}. \tag{6}$$

Substitute (5) into (6), we obtain

$$\Delta\theta_{\text{a}} = \begin{cases} -\dfrac{1}{45}\theta_{\text{guide}}^2 + 5\theta_{\text{guide}} - 202.5, & \theta_{\text{guide}} \in (90, 135) \\ 202.5 - \theta_{\text{guide}}, & \theta_{\text{guide}} \in (135, 180). \end{cases} \tag{7}$$

The $\Delta\theta_{\text{a}} - \theta_{\text{guide}}$ curve is illustrated in Fig. 7(b), revealing that when $\theta_{\text{guide}}$ is in the horizontal direction, its average prediction error relative to the ground-truth region is the smallest, which is the optimal guidance direction.

In conclusion, the essence of the optimal guiding direction is that with the minimum mean error, the average error is calculated by intercepting a part of the error curve through an interception window, sliding along the error curve with $\theta_{\text{guide}}$, and calculating its integral. The period of the error curve is 90° (because the period of the included angle between two rectangle boxes with a large aspect ratio is 90°), and the size of the interception window is 45°. When $\theta_{\text{guide}}$ reaches the horizontal direction, the interception window is located at the smallest part of the error curve within a period. The average error obtained by the integral operation is minimized, and the horizontal direction is the optimal guidance direction. In addition, the above method treats codes "00000" and "11111" as equivalent encodings, thus avoiding boundary problems. The flowchart and pseudocode of DAL are presented in Fig. 8 and Algorithm 25, respectively.

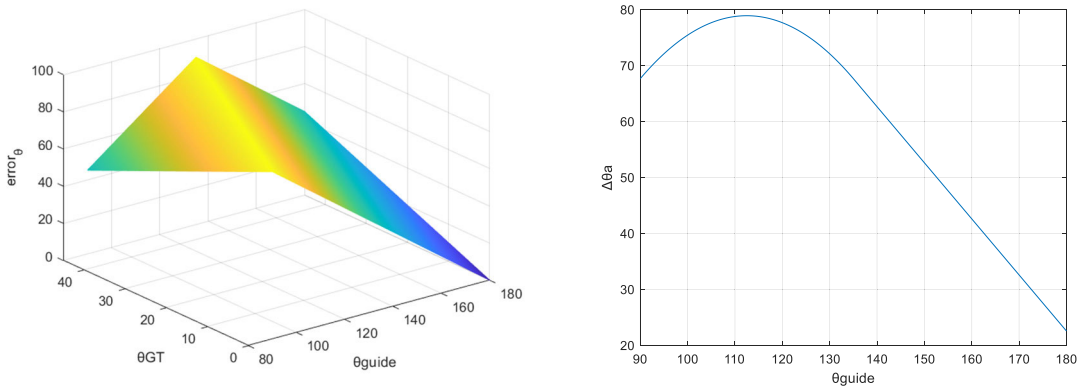
Yang et al. [46] pointed out that BCL's low tolerance may cause a significant variability of the adjacent angles to the code, thus leading to no positive correlation between the coding loss and the actual error. Thus, the angle distance is unknown and cannot be adequately measured by the loss. DAL treats angle learning as a converging and progressive process, focusing on guiding the direction of the binary search rather than strictly punishing the prediction coding bits. Assuming that the ground-truth label is "01011," the predicted code is "00011," and the first error bit is at $b_2$. Then, DAL will dynamically adjust "011" after $b_2$ of the ground-truth label to "111." During training, a punitive loss is calculated for the $b_2$ bit, and the guiding loss will be calculated for the bits after the $b_2$. This indicates that the purpose of DAL is guiding the subsequent coding bits to converge to the minimal error direction based on the $b_2$ bit prediction error. This convergence is regional and irrelevant, with the coding distance in the region being knowable.

DAL has no strict dependence on the angle distance but possesses the characteristic that the smaller the first error bit's granularity, the smaller the maximum error. Hence, DAL avoids being influenced by the unknown angle distance problem in BCL.

### C. Gradient Truncation Mechanism

Since each angle coding bit corresponds to a binary classification task, we select the cross-entropy loss for training

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(a) (b)

Fig. 7. Principle of guidance error: (a) distribution of guidance error and (b) $\Delta\theta_{\mathrm{a}} - \theta_{\mathrm{guide}}$ curve.

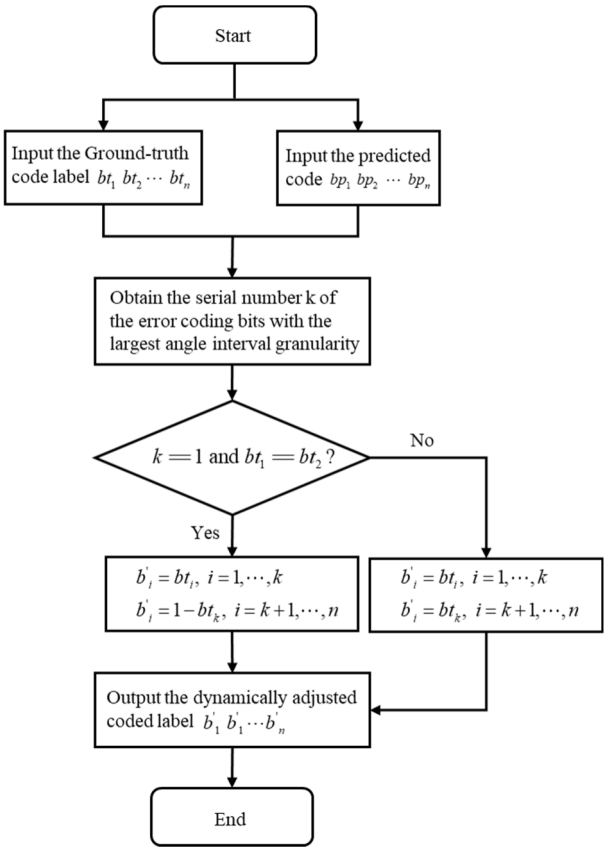



Fig. 8. DAL flowchart.

the coding layers, defined as

$$L_{\mathrm{b}}\left(\mathrm{bt}_i, \mathrm{bp}_i\right) = -\mathrm{bt}_i \cdot \log\left(\mathrm{bp}_i\right) - \left(1 - \mathrm{bt}_i\right) \cdot \log\left(1 - \mathrm{bp}_i\right) \tag{8}$$

where $\mathrm{bt}_i$ and $\mathrm{bp}_i$ are the $i$th ground truth and predicted bits, respectively.

We add a gradient truncation mechanism to the cross-entropy loss to prevent the model from over-optimizing the simple samples and ignoring the complex samples during training. When the network prediction of the coding bit reaches the threshold value, its loss is set to 0 by the truncation function to block the gradient backpropagation and avoid over-fitting. The truncation function is set as

$$T(x) = \frac{\mathrm{sgn}(x) + 1}{2} \tag{9}$$

where $\mathrm{sgn}(\cdot)$ is the sign function. Moreover, the function's curve is shown in Fig. 9.

**Algorithm 2** Dynamic Adjusting Labels (DAL) Strategy

**Input:** prediction code $L_P$, ground-truth code label $L_{GT}$.
**Output:** dynamically adjusted label $L_{DAL}$.

```
L_DAL = []
n = length(L_GT)
# Adjust labels.
for i in range(n) do
    # Determine whether each coding bit of L_P and L_GT are equal. If yes, adjust the labels and jump out of the loop. Otherwise, copy the coding bit of L_GT to the same position of L_DAL.
    if abs(L_P[i]- L_GT[i]) > 0 then
        # Determine whether θ_GT is in subinterval I or subinterval IV and the prediction of b_1 is wrong.
        if i = 1 and L_GT[1] = L_GT[2] then
            # Adjust labels to horizontal detection.
            for j in range(n) do
                L_DAL.append(L_P[1])
            end
        else
            # Adjust labels to the optimal guidance direction.
            L_DAL.append(L_P[i])
            for j in range(n-i) do
                L_DAL.append(L_GT[i])
            end
        end
        # Jump out of the loop.
        break
    end
    L_DAL.append(L_GT[i])
end
Return L_DAL
```

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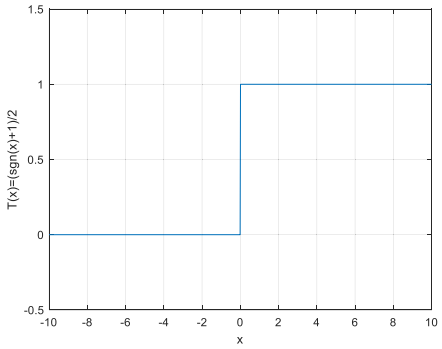



Fig. 9. Truncation function curve.

After adding the truncation function, the coding bit loss becomes

$$\begin{aligned} &L_{\mathrm{b}}\left(\mathrm{bt}_i, \mathrm{bp}_i\right) \\ &= -\mathrm{bt}_i \cdot \log\left(\mathrm{bp}_i\right) \cdot T\left(\mathrm{thr} - \mathrm{bp}_i\right) \\ &\quad - \left(1 - \mathrm{bt}_i\right) \cdot \log\left(1 - \mathrm{bp}_i\right) \cdot T\left(\mathrm{bp}_i + \mathrm{thr} - 1\right) \end{aligned} \tag{10}$$

where thr is the truncation threshold. When the positive sample's prediction exceeds thr, or the negative sample's prediction is less than $1 -$ thr, the truncation mechanism is triggered, and the gradient is set to 0.

By coding the bit loss, we further obtain the angle loss as follows:

$$\begin{aligned} L_\theta &= \frac{1}{n}\sum_{i=1}^{n} L_{\mathrm{b}}\left(\mathrm{bt}_i, \mathrm{bp}_i\right) \\ &= -\frac{1}{n}\sum_{i=1}^{n} \begin{bmatrix} \mathrm{bt}_i \cdot \log\left(\mathrm{bp}_i\right) \cdot T\left(\mathrm{thr} - \mathrm{bp}_i\right) \\ +\left(1 - \mathrm{bt}_i\right) \cdot \log\left(1 - \mathrm{bp}_i\right) \cdot T\left(\mathrm{bp}_i + \mathrm{thr} - 1\right) \end{bmatrix}. \end{aligned} \tag{11}$$

### D. Loss Function

In order to balance performance and speed, we choose Yolov5m as the basic model and implement the proposed method. The bounding box parameter regression formula is

$$\begin{aligned} &b_x = 2\sigma\left(t_x\right) - 0.5 + c_x, \quad b_y = 2\sigma\left(t_y\right) - 0.5 + c_y \\ &b_{\mathrm{w}} = p_{\mathrm{w}} \cdot \left[2\sigma\left(t_{\mathrm{w}}\right)\right]^2, \quad b_{\mathrm{h}} = p_{\mathrm{h}} \cdot \left[2\sigma\left(t_{\mathrm{h}}\right)\right]^2 \\ &\hat{b}_x = 2\sigma\left(\hat{t}_x\right) - 0.5 + c_x, \quad \hat{b}_y = 2\sigma\left(\hat{t}_y\right) - 0.5 + c_y \\ &\hat{b}_{\mathrm{w}} = p_{\mathrm{w}} \cdot \left[2\sigma\left(\hat{t}_{\mathrm{w}}\right)\right]^2, \quad \hat{b}_{\mathrm{h}} = p_{\mathrm{h}} \cdot \left[2\sigma\left(\hat{t}_{\mathrm{h}}\right)\right]^2 \end{aligned} \tag{12}$$

where $\hat{b}_x$, $\hat{b}_y$, $\hat{b}_{\mathrm{w}}$, and $\hat{b}_{\mathrm{h}}$ are the central coordinates and dimensions of the ground-truth bounding box, $b_x$, $b_y$, $b_{\mathrm{w}}$, and $b_{\mathrm{h}}$ are the central coordinates and dimensions of the predicted bounding box, and $c_x$ and $c_y$ are the top-left coordinates of the grid where the anchor is located. $p_{\mathrm{w}}$ and $p_{\mathrm{h}}$ indicate the anchor's width and height, respectively, and $\sigma(\cdot)$ is the activation function used to map the offset values of the bounding box center coordinates relative to the grid top-left corner between 0 and 1. Yolov5m guides the training process of the bounding box parameters through the complete IoU (CIoU) [66] indicator, which is calculated as follows:

$$\begin{cases} \mathrm{CIoU} = \mathrm{IoU} - \dfrac{\rho^2}{c^2} - \alpha\upsilon \\ \rho^2 = \left(\hat{b}_x - b_x\right)^2 + \left(\hat{b}_y - b_y\right)^2 \\ c^2 = \left[\max\left(\hat{b}_x + \dfrac{\hat{b}_{\mathrm{w}}}{2}, b_x + \dfrac{b_{\mathrm{w}}}{2}\right) - \min\left(\hat{b}_x - \dfrac{\hat{b}_{\mathrm{w}}}{2}, b_x - \dfrac{b_{\mathrm{w}}}{2}\right)\right]^2 \\ \quad + \left[\max\left(\hat{b}_y + \dfrac{\hat{b}_{\mathrm{h}}}{2}, b_y + \dfrac{b_{\mathrm{h}}}{2}\right) - \min\left(\hat{b}_y - \dfrac{\hat{b}_{\mathrm{h}}}{2}, b_y - \dfrac{b_{\mathrm{h}}}{2}\right)\right]^2 \\ \upsilon = \dfrac{4}{\pi^2}\left(\arctan\dfrac{\hat{b}_{\mathrm{w}}}{\hat{b}_{\mathrm{h}}} - \arctan\dfrac{b_{\mathrm{w}}}{b_{\mathrm{h}}}\right)^2 \\ \alpha = \dfrac{\upsilon}{(1 - \mathrm{IoU}) + \upsilon} \end{cases} \tag{13}$$

where IoU, $\rho$, and $c$ denote the intersection over union, central Euclidean distance, and diagonal length of the minimum enclosing rectangle in the union area between the predicted and ground-truth bounding boxes, respectively. $\upsilon$ is used to measure the aspect ratio consistency of the two bounding boxes, and $\alpha$ is a positive tradeoff coefficient.

We employ the CIoU loss [66] as the box's regression loss

$$L_{\mathrm{box}} = \frac{1}{N}\sum_{i}^{N}\left[1 - \mathrm{CIoU}\left(\hat{b}_{ix}, \hat{b}_{iy}, \hat{b}_{iw}, \hat{b}_{ih}, b_{ix}, b_{iy}, b_{iw}, b_{ih}\right)\right] \tag{14}$$

where $N$ is the number of anchors.

Both the classification and confidence losses adopt the cross-entropy loss, as presented in the following equations:

$$L_{\mathrm{cls}} = \frac{1}{N}\sum_{i}^{N}\left[\hat{c}_i \log(c_i) + \left(1 - \hat{c}_i\right)\log(1 - c_i)\right] \tag{15}$$

$$L_{\mathrm{obj}} = \frac{1}{N}\sum_{i}^{N}\left[\hat{o}_i \log(o_i) + \left(1 - \hat{o}_i\right)\log(1 - o_i)\right] \tag{16}$$

where $\hat{c}$ and $c$ represent the ground-truth label and the prediction probability of each category, respectively, and $\hat{o}$ and $o$ are the ground-truth label and foreground prediction scores, respectively.

Based on Yolov5m, the multitask loss function in this article adds the angle loss, defined as

$$L = \lambda_{\mathrm{box}} L_{\mathrm{box}} + \lambda_{\mathrm{cls}} L_{\mathrm{cls}} + \lambda_{\mathrm{obj}} L_{\mathrm{obj}} + \lambda_\theta \frac{1}{N}\sum_{i}^{N} L_\theta \tag{17}$$

where $\lambda_{\mathrm{box}}$, $\lambda_{\mathrm{cls}}$, $\lambda_{\mathrm{obj}}$, and $\lambda_\theta$ are the weight coefficients per loss, set as {0.25, 3.0, 2.0, 0.5} by default.

## IV. Experiments and Results

This section evaluates the proposed method on public datasets. Specifically, first, we introduce the ship datasets FGSD2021 [67], HRSC2016 [68], and DOTA-ship. Then, we perform ablation experiments and challenge our method against state-of-the-art (SOTA) methods. Finally, we analyze and discuss the effects of the proposed method on precision

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TABLE I
THREE LEVELS OF DETECTION TASKS OF HRSC2016

| $L_1$ | $L_2$ |
|---|---|
| Ship | Aircraft carrier (Air.) |
| | Warship (War.) |
| | Merchant ship (Mer.) |
| | Submarine (Sub.) |

distribution and angle learning. The optimal and suboptimal results in the experiments are marked in red and blue, respectively, and are shown in bold. The results shown only in bold indicate the optimal results.

### A. Datasets

*1) FGSD2021:* This dataset includes Google Earth high-resolution remote sensing images and some images from the HRSC2016 dataset, with the resolution normalized to 1 m per pixel. The image width ranges from 157 to 7789, and the height ranges from 224 to 6506. The dataset is divided into 424 training images and 212 test images containing 3909 and 1849 target instances, respectively. There are 20 categories annotated, including Aircraft carrier (Air.), War ship (War.), Tarawa class (Tar.), Austin class (Aus.), Whidbey Island class (Whi.), San Antonio class (San.), Newport class (New.), Ticonderoga class (Tic.), Arleigh Burke class (Bur.), Perry class (Per.), Lewis and Clark class (Lew.), Supply class (Sup.), Henry J. Kaiser class (Kai.), Bob Hope class (Hop.), Mercy class (Mer.), Freedom class (Fre.), Independence class (Ind.), Avenger class (Ave.), Submarine (Sub.), and Other (Oth.). All targets are marked by a rotated box in the format of (cx, cy, w, h, angle).

*2) HRSC2016:* The experiments utilize the public ship detection dataset HRSC2016 [68], collected from Google Earth and published by Northwestern Polytechnical University, Xi'an, China, in 2016, including annotated oriented bounding boxes (OBBs). The dataset covers near-shore and offshore scenes, including 1061 images with resolutions ranging from 0.4 to 2 m and sizes from 300 × 300 to 1500 × 90 pixels. The training, validation, and test sets include 436, 181, and 444 images with 1207, 541, and 1228 samples, respectively, as each image may contain more than one ship. The detection tasks of HRSC2016 comprise three levels: $L_1$, $L_2$, and $L_3$, where $L_1$ involves only one class, $L_2$ has four classes, and $L_3$ involves 27 classes. The details of the detection tasks are reported in Table I. The single and multicategory object experiments utilize the $L_1$ and $L_2$ tasks. Considering that this article does not focus on a fine-grained classification task, we believe that the multicategory detection through the $L_2$ task is sufficient, and thus, the $L_3$ task is neglected. *3) DOTA-Ship:* DOTA [69] is a typical remote sensing dataset in object detection. It contains 2806 images with an image size ranging from 800 × 800 to 4000 × 4000 pixels, containing 188 282 instances labeled with arbitrary quadrilaterals. The dataset involves 15 categories. We extract the ship class to obtain the dataset DOTA-ship, and since the test set of DOTA does not contain labels, we only extract its training and validation sets to build the DOTA-ship. The training set of DOTA-ship contains 326 images and 28 068 instances, and the validation set has 108 images and 8960 instances.

### B. Implementation Details

The proposed method is implemented in Anaconda Pytorch 1.7.1 + CUDA 10.2 based on Ubuntu 20.04, and all our experiments are conducted on a workstation with an Intel[1] Xeon[1] Gold 6128 CPU and an NVIDIA Quadro RTX 5000 (16-GB graphics memory) GPU. We use the stochastic gradient descent (SGD) [70] method as the optimizer to train the network for 200 epochs with a batch size of 16, and the learning rate is planned by cosine annealing [71]. The initial learning rate, weight decay, and momentum are set to 0.002, 0.0005, and 0.937, respectively. We crop the images of FGSD2021 and DOTA-ship into 800 × 800 pixels subimages with an overlap of 204 pixels. Additionally, the FGSD2021 subimages are scaled to 1024 × 1024 pixels in the training and testing stages. Since the image size of the HRSC2016 dataset is small, we directly resize it to 800 × 800 pixels without clipping it. In the training stage, we used random flipping, random spinning, mosaic [33], and mixup [72] for data enhancement. In the test stage, we combine the detection results of the FGSD2021 and DOTA-ship subimages. Then, we delete the repetitive detections using rotated-nonmaximum suppression (RNMS) and obtain the detection results of the two datasets.

### C. Evaluation Metrics

The positive and negative results are determined based on the IoU between the two boxes, while the precision–recall curve (PRC) and the mean average precision (mAP) are used to evaluate the detectors' performance.

*1) IoU:* IoU is the ratio of two polygons' overlapping area to their union area. In this work, we use the IoU to measure the overlap degree between the predicted and ground-truth bounding boxes. If the IoU is higher than a threshold, the detection block is accurately positioned and marked as a true positive (TP). Otherwise, it is marked as a false positive (FP). False negative (FN) is marked for a ground-truth bounding box that does not match a prediction.

*2) PRC:* The precision and recall are defined as follows:

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{18}$$

$$\text{recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{19}$$

where precision refers to the proportion of TP samples in the predicted positive samples, and recall refers to the proportion of TP samples in ground-truth positive samples. After sorting all the predicted samples according to the confidence degree, we calculate the corresponding precision and recall from top-1 to top-$n$ and then take the results as the coordinates of the 2-D plane points to draw the PRC.

[1]Registered trademark.

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TABLE II
PERFORMANCE OF DAL ACHIEVED ON THREE DATASETS

| Method | n | HRSC2016 $L_1$ | | | | FGSD2021 | | | | DOTA-ship | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | $mAP_{50}$ (07/12) | $mAP_{75}$ (07/12) | $mAP_{95}$ (07/12) | $mAP_{50:95}$ (07/12) | $mAP_{50}$ (07/12) | $mAP_{60}$ (07/12) | $mAP_{70}$ (07/12) | $mAP_{80}$ (07/12) | $mAP_{50}$ (07/12) | $mAP_{60}$ (07/12) | $mAP_{70}$ (07/12) | $mAP_{80}$ (07/12) |
| BCL [52] | 3 | 87.9/92.2 | 47.5/57.2 | 3.0/2.8 | 48.4/55.3 | 83.4/88.3 | 71.6/78.2 | 49.1/58.8 | 25.8/34.8 | 89.0/95.5 | 87.4/91.5 | 41.6/53.6 | 10.5/8.4 |
| | 4 | 89.5/95.1 | 73.3/79.8 | 2.3/4.8 | 63.8/70.0 | 87.2/92.8 | 86.4/91.8 | 80.0/85.6 | 38.6/47.3 | **89.4**/95.9 | 87.8/92.4 | 62.5/71.2 | 19.4/22.6 |
| | 5 | 89.6/95.4 | 79.7/90.8 | 6.2/11.4 | 71.2/78.3 | 88.3/93.5 | 88.1/93.0 | 84.9/89.8 | 67.2/74.9 | **89.4**/96.2 | 87.6/92.1 | 63.0/71.9 | 21.8/27.4 |
| | 6 | 89.4/95.1 | 78.8/89.6 | 9.1/15.7 | 71.5/78.9 | 88.0/92.2 | 87.4/91.6 | 83.5/88.6 | 66.5/74.3 | 89.2/95.8 | 87.5/91.9 | 63.5/72.4 | 21.2/25.9 |
| | 7 | 89.5/95.1 | 79.2/90.5 | 6.2/12.7 | 71.0/78.2 | 86.8/92.0 | 86.3/91.2 | 81.5/87.1 | 67.1/74.1 | 89.0/95.5 | 87.1/91.3 | 62.5/71.2 | 20.5/25.0 |
| | 8 | 89.5/95.6 | 77.7/87.9 | 6.7/11.2 | 69.4/76.0 | 87.4/92.7 | 86.2/91.5 | 80.7/86.9 | 62.5/71.2 | 89.1/95.6 | 86.7/90.6 | 61.7/69.4 | 18.3/23.4 |
| BCL+DAL | 3 | 87.5/91.8 | 48.6/58.4 | 4.6/2.9 | 48.7/55.3 | 83.8/89.1 | 71.8/78.4 | 50.9/59.5 | 25.6/34.3 | **89.4**/**96.2** | 87.7/92.1 | 40.7/52.1 | 10.5/8.0 |
| | 4 | **89.7**/96.0 | 75.8/84.0 | 3.0/5.2 | 64.3/71.4 | 89.3/93.8 | 87.5/92.5 | 80.0/85.9 | 36.3/45.7 | **89.4**/96.1 | **88.1**/**92.9** | 63.8/73.2 | 19.9/23.9 |
| | 5 | 89.6/**96.1** | 87.5/91.6 | 5.1/10.7 | 71.8/78.4 | 89.7/93.5 | 88.6/93.0 | 85.8/90.7 | 68.1/75.6 | 88.9/95.4 | 87.5/91.9 | **63.9**/**73.3** | **22.8**/**29.3** |
| | 6 | 89.4/95.4 | 87.6/91.5 | **15.0**/**16.2** | **74.5**/**80.7** | 89.1/93.8 | 88.4/93.2 | 85.9/90.7 | **70.3**/**77.9** | 88.6/95.2 | 87.2/91.6 | 63.8/72.8 | 22.0/27.5 |
| | 7 | 89.5/95.6 | **87.7**/**91.8** | 14.1/14.8 | 74.0/80.2 | 89.9/**94.4** | **89.8**/**94.2** | **86.6**/**91.1** | 70.0/76.8 | 89.2/95.8 | 87.9/92.6 | 63.6/73.1 | 21.9/27.9 |
| | 8 | 89.5/96.0 | 86.9/91.0 | 11.1/15.3 | 74.3/80.5 | **90.0**/94.1 | 87.9/93.2 | 84.8/90.3 | 67.3/75.1 | 89.0/95.4 | 87.2/91.4 | 63.2/71.8 | 21.6/26.9 |

TABLE III
PERFORMANCE OF DIFFERENT THR ON $L_2$ TASK OF HRSC2016

| n | *thr* 0.6 | 0.7 | 0.8 | 0.9 | $mAP_{50}$ (12) | $mAP_{75}$ (12) | $mAP_{95}$ (12) | $mAP_{50:95}$ (12) |
|---|---|---|---|---|---|---|---|---|
| 3 | √ | | | | 83.4 | 57.3 | 1.9 | **52.3** |
| | | √ | | | 83.4 | **58.5** | 1.9 | 52.2 |
| | | | √ | | 83.3 | 57.3 | **2.0** | 52.2 |
| | | | | √ | **84.6** | 57.6 | 1.5 | 51.8 |
| 4 | √ | | | | 88.3 | 70.8 | **4.6** | 63.3 |
| | | √ | | | **89.6** | **74.7** | 3.7 | **65.1** |
| | | | √ | | 89.3 | 74.6 | 4.1 | 64.3 |
| | | | | √ | 89.3 | 74.7 | 4.2 | 64.5 |
| 5 | √ | | | | 89.1 | 82.5 | 8.3 | 70.4 |
| | | √ | | | **89.5** | 82.1 | **8.6** | **71.3** |
| | | | √ | | 89.3 | 83.3 | 8.5 | 70.9 |
| | | | | √ | 89.0 | **83.4** | 8.2 | 70.3 |
| 6 | √ | | | | 89.4 | 81.4 | 9.5 | 70.4 |
| | | √ | | | **90.1** | **81.5** | 8.5 | **71.3** |
| | | | √ | | 89.7 | 77.5 | **10.0** | 70.0 |
| | | | | √ | 88.4 | 80.4 | 7.9 | 69.5 |
| 7 | √ | | | | 89.7 | 80.3 | 7.0 | 69.1 |
| | | √ | | | **89.9** | 81.8 | **8.3** | **70.7** |
| | | | √ | | 89.5 | 80.0 | 6.3 | 69.6 |
| | | | | √ | 88.5 | **83.2** | 6.8 | 70.0 |
| 8 | √ | | | | 89.0 | 72.6 | 4.8 | 64.5 |
| | | √ | | | **89.9** | **80.0** | **6.8** | **69.3** |
| | | | √ | | 89.0 | 79.5 | 6.4 | 68.4 |
| | | | | √ | 89.4 | 79.8 | 5.8 | 69.0 |

TABLE IV
PERFORMANCE OF GRADIENT TRUNCATION MECHANISM ACHIEVED ON $L_2$ TASK OF HRSC2016

| Method | n | T($x$) | Air. | War. | Mer. | Sub. | $mAP_{50}$ (07/12) | $mAP_{75}$ (07/12) | $mAP_{95}$ (07/12) | $mAP_{50:95}$ (07/12) |
|---|---|---|---|---|---|---|---|---|---|---|
| BCL [52] | 3 | | **88.1**/**92.4** | 84.5/89.0 | 70.6/77.9 | 54.5/68.2 | 74.4/81.9 | 46.5/57.2 | **1.6**/1.7 | 42.6/50.4 |
| | | √ | 85.9/90.0 | **85.9**/**90.7** | **72.1**/**79.3** | **63.9**/**73.5** | **77.0**/**83.4** | **48.9**/**58.5** | 0.8/**1.9** | **45.6**/**52.2** |
| | 4 | | 78.2/90.2 | 87.1/93.6 | 69.6/76.5 | **81.8**/**92.2** | 79.2/88.2 | 61.1/71.7 | 1.2/3.2 | 54.0/62.7 |
| | | √ | **86.5**/**90.8** | **87.3**/**94.0** | **73.2**/**81.9** | **81.8**/91.7 | **82.2**/**89.6** | **64.7**/**74.7** | **1.6**/**3.7** | **57.1**/**65.1** |
| | 5 | | 79.2/90.3 | 87.7/94.5 | 70.9/**78.4** | **80.9**/91.0 | 79.7/88.5 | 74.0/81.8 | **4.9**/7.7 | 61.9/70.6 |
| | | √ | **87.6**/**93.5** | **87.8**/**94.6** | **72.0**/78.3 | **80.9**/**91.5** | **82.1**/**89.5** | **74.5**/**82.1** | 4.6/**8.6** | **63.7**/**71.3** |
| | 6 | | **87.9**/**93.1** | 87.0/93.6 | 72.3/79.9 | 72.7/86.3 | 80.0/88.3 | 72.4/80.6 | **5.5**/7.9 | 60.8/68.9 |
| | | √ | 87.5/92.5 | **87.3**/**93.8** | **73.8**/**81.9** | **81.8**/**92.2** | **82.6**/**90.1** | **73.8**/**81.5** | 5.0/**8.5** | **63.7**/**71.3** |
| | 7 | | 86.2/90.4 | **87.5**/**94.4** | 70.3/78.1 | 80.7/89.2 | 81.2/88.0 | 71.9/80.6 | 5.3/5.7 | 61.2/68.8 |
| | | √ | **87.3**/**91.6** | **87.5**/93.9 | **74.5**/**82.9** | **81.1**/**91.2** | **82.6**/**89.9** | **74.5**/**81.8** | **6.4**/**8.3** | **63.6**/**70.7** |
| | 8 | | **88.1**/**92.5** | 87.5/94.5 | 72.4/80.0 | 71.4/86.5 | 79.8/88.4 | 69.2/78.3 | **7.3**/**7.2** | 59.9/67.7 |
| | | √ | 78.3/90.1 | **87.7**/**94.7** | **73.0**/**81.6** | **81.6**/**93.3** | **80.1**/**89.9** | **72.8**/**80.0** | 5.3/6.8 | **61.3**/**69.3** |

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TABLE V
AP OF DIFFERENT OBJECTS ON FGSD2021

| Method | n | Air. | Was. | Tar. | Aus. | Whi. | San. | New. | Tic. | Bur. | Per. | Lew. | Sup. | Kai. | Hop. | Mer. | Fre. | Ind. | Ave. | Sub. | Oth. | mAP(07) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $R^3$Det [73] | - | 90.9 | 80.9 | 81.5 | 90.1 | 79.3 | 87.5 | 29.5 | 77.4 | 89.4 | 69.7 | 59.9 | 67.3 | 80.7 | 76.8 | 72.7 | 83.3 | **90.9** | 38.4 | 23.1 | 40.0 | 70.47 |
| RetinaNet-Rbb [50] | - | 89.7 | 89.2 | 78.2 | 87.3 | 77.0 | 86.9 | 62.7 | 81.5 | 83.3 | 70.6 | 46.8 | 69.9 | 80.2 | 83.1 | **100** | 80.6 | 89.7 | 61.5 | 42.5 | 9.1 | 73.49 |
| CSL [45] | - | 89.7 | 81.3 | 77.2 | 80.2 | 71.4 | 77.2 | 52.7 | 87.7 | 87.7 | 74.2 | 57.1 | 97.2 | 77.6 | 80.5 | **100** | 72.7 | **100** | 32.6 | 37.0 | 40.7 | 73.73 |
| RSDet [43] | - | 89.8 | 80.4 | 75.8 | 77.3 | 78.6 | 88.8 | 26.1 | 84.7 | 87.6 | 75.2 | 55.1 | 74.4 | 89.7 | 89.3 | **100** | 86.4 | **100** | 27.6 | 37.6 | 50.6 | 73.74 |
| SCRDet [42] | - | 77.3 | 90.4 | 87.4 | 89.8 | 78.8 | **90.9** | 54.5 | 88.3 | 89.6 | 74.9 | 68.4 | 59.2 | 90.4 | 77.2 | 81.8 | 73.9 | **100** | 43.9 | 43.8 | 57.1 | 75.90 |
| $R^2$CNN [60] | - | 89.9 | 80.9 | 80.5 | 79.4 | 87.0 | 87.8 | 44.2 | 89.0 | 89.6 | 79.5 | 80.4 | 47.7 | 81.5 | 87.4 | **100** | 82.4 | **100** | 66.4 | 50.9 | 57.2 | 78.09 |
| $S^2$A-Net [63] | - | 90.9 | 81.4 | 73.3 | 89.1 | 80.9 | 89.9 | 81.2 | 89.2 | 90.7 | 88.9 | 60.5 | 75.9 | 81.6 | 89.2 | **100** | 68.6 | **90.9** | 61.3 | 55.7 | 64.7 | 80.19 |
| DARDet [75] | - | 90.9 | 89.2 | 69.7 | 89.6 | 88.0 | 81.4 | 90.3 | 89.5 | 90.5 | 79.7 | 62.5 | 87.9 | 90.2 | 89.2 | **100** | 68.9 | 81.8 | 66.3 | 44.3 | 56.2 | 80.31 |
| Oriented R-CNN [76] | - | 90.9 | 89.7 | 81.5 | 81.1 | 79.6 | 88.2 | **98.9** | 89.8 | 90.6 | 87.8 | 60.4 | 73.9 | 81.8 | 86.7 | **100** | 60.0 | **100** | 79.4 | 66.9 | 63.7 | 82.54 |
| ROI-Trans [40] | - | 90.9 | 88.6 | 87.2 | 89.5 | 78.5 | 88.8 | 81.8 | 89.6 | 89.8 | 90.4 | 71.7 | 74.7 | 73.7 | 81.6 | 78.6 | **100** | 75.6 | 78.4 | 68.0 | 66.9 | 83.48 |
| BBAVectors [74] | - | 99.5 | **90.9** | 75.9 | **94.3** | **90.9** | 52.9 | 88.5 | 90.0 | 80.4 | 72.2 | 76.9 | 88.2 | **99.6** | **100** | 94.0 | **100** | 74.5 | 58.9 | 63.1 | **81.1** | 83.59 |
| ReDet [77] | - | 90.9 | 90.6 | 80.3 | 81.5 | **89.3** | 88.4 | 81.8 | 88.8 | 90.3 | 90.5 | 78.1 | 76.0 | 90.7 | 87.0 | **98.2** | 84.4 | **90.9** | 74.6 | **85.3** | **81.2** | 85.44 |
| CHPDet [67] | - | 90.9 | 90.2 | 90.9 | 90.3 | **89.3** | 89.2 | **98.9** | 90.2 | 90.2 | 90.2 | 72.2 | 96.5 | 90.7 | 95.3 | **100** | 95.2 | **90.9** | 86.4 | **85.9** | 62.4 | 89.29 |
| Baseline | 3 | 85.1 | **90.9** | 89.6 | 90.1 | 87.1 | **90.7** | 94.3 | 71.8 | **90.9** | 87.9 | 77.4 | **100** | 79.4 | 77.3 | 96.4 | 93.9 | **90.9** | **87.1** | 17.8 | 70.1 | 83.43 |
| | 4 | 90.4 | **90.9** | 96.8 | **90.5** | 87.9 | 89.2 | **96.0** | 89.6 | **90.8** | **90.9** | 76.8 | 87.9 | 88.8 | 82.0 | 81.8 | 97.0 | **90.9** | 83.8 | 66.3 | 75.0 | 87.17 |
| | 5 | 89.6 | **90.9** | **97.9** | 90.1 | 88.3 | 89.3 | 95.5 | 89.8 | **90.9** | **90.8** | 76.4 | 97.9 | 78.1 | 86.2 | 97.3 | 97.0 | **90.9** | 84.9 | 70.4 | 74.6 | 88.33 |
| | 6 | 83.6 | **90.9** | **97.9** | 90.2 | 86.3 | 86.8 | 95.5 | 89.6 | **90.8** | 90.7 | 74.8 | 97.2 | 80.5 | 79.2 | **100** | 96.4 | **100** | 81.9 | 74.6 | 73.2 | 87.99 |
| | 7 | 83.2 | **90.8** | 90.3 | 90.4 | 86.1 | 90.3 | 95.5 | 90.3 | 90.5 | 90.5 | 68.0 | 94.9 | 79.8 | 82.8 | 91.8 | 97.0 | **100** | 81.3 | 69.9 | 73.5 | 86.83 |
| | 8 | 86.5 | 90.7 | 97.3 | 90.0 | 86.6 | **90.7** | 94.9 | 89.8 | **90.8** | **90.9** | 68.9 | 89.4 | 80.8 | 85.1 | **98.2** | 93.9 | **100** | 82.4 | 66.9 | 73.8 | 87.37 |
| Our Method | 3 | 90.9 | **90.9** | 97.2 | 90.1 | 87.8 | **90.7** | 85.5 | 73.3 | 90.8 | 87.5 | 78.3 | 89.4 | 81.3 | 72.3 | 97.3 | 96.4 | **90.9** | 86.4 | 18.9 | 64.3 | 83.01 |
| | 4 | 89.8 | **90.9** | 89.7 | 90.0 | 87.8 | 90.1 | 85.2 | 99.1 | **90.9** | **90.9** | **79.1** | **98.7** | **97.2** | 94.6 | 95.5 | **98.2** | **90.9** | 84.1 | 84.3 | 75.5 | 90.12 |
| | 5 | 88.8 | **90.9** | 97.1 | 90.2 | 85.9 | 88.3 | 94.9 | 99.0 | **90.8** | 90.4 | **78.8** | 77.7 | 88.2 | 85.0 | 94.6 | 94.6 | **100** | 86.3 | 80.9 | 74.8 | 88.85 |
| | 6 | **99.8** | **90.9** | **97.9** | 90.3 | 88.1 | 88.7 | 94.9 | 90.0 | 90.6 | **90.9** | 77.3 | 97.9 | 79.8 | 81.5 | **100** | 97.6 | **100** | **92.3** | 81.8 | 75.6 | **90.29** |
| | 7 | **97.3** | **90.9** | **98.1** | **90.5** | 87.3 | **90.7** | 94.9 | **99.1** | 90.7 | **90.9** | 69.0 | 96.4 | 87.4 | 97.7 | **98.2** | 96.4 | **100** | 76.0 | 73.5 | 74.9 | 89.98 |
| | 8 | 90.1 | **90.9** | 96.2 | 89.9 | 87.2 | 90.3 | 95.5 | **99.2** | **90.8** | 90.4 | 78.2 | 97.9 | 87.3 | **98.4** | **100** | 95.8 | **90.9** | 85.8 | 79.8 | 75.6 | **90.50** |

TABLE VI
DETECTION ACCURACY WITH THE SOTA METHODS ON FGSD2021

| Method | Backbone | n | $mAP_{50}(07)$ | $mAP_{60}(07)$ | $mAP_{70}(07)$ | $mAP_{80}(07)$ |
|---|---|---|---|---|---|---|
| $R^3$Det [73] | ResNet50 | - | 70.47 | 68.32 | 57.17 | 27.44 |
| RetinaNet-Rbb [50] | ResNet50 | - | 73.49 | 69.17 | 62.82 | 45.00 |
| CSL [45] | ResNet50 | - | 73.73 | 69.71 | 60.25 | 34.93 |
| RSDet [43] | ResNet50 | - | 73.74 | 69.55 | 61.52 | 35.83 |
| SCRDet [42] | ResNet50 | - | 75.90 | 70.98 | 61.82 | 35.12 |
| $R^2$CNN [60] | ResNet50 | - | 78.09 | 75.03 | 64.83 | 36.41 |
| $S^2$A-Net [63] | ResNet50 | - | 80.19 | 79.58 | 75.65 | 58.52 |
| DARDet [75] | ResNet50 | - | 80.31 | 79.62 | 74.77 | 59.21 |
| Oriented R-CNN [76] | ResNet50 | - | 82.54 | 81.32 | 78.53 | 64.87 |
| ROI-Trans [40] | ResNet50 | - | 83.48 | 82.63 | 80.55 | 65.18 |
| BBAVectors [74] | ResNet50 | - | 83.59 | 82.74 | 78.55 | 62,48 |
| ReDet [77] | ReResNet50 | - | 85.44 | 84.65 | 80.24 | 67.94 |
| CHPDet [67] | DLA34 | - | 89.29 | 88.98 | **86.57** | **73.56** |
| Baseline | CSPDarknet53 | 3 | 83.43 | 71.62 | 49.13 | 25.77 |
| | | 4 | 87.17 | 86.36 | 80.03 | 38.60 |
| | | 5 | 88.33 | 88.06 | 84.86 | 67.17 |
| | | 6 | 87.99 | 87.40 | 83.54 | 66.50 |
| | | 7 | 86.83 | 86.32 | 81.47 | 67.05 |
| | | 8 | 87.37 | 86.17 | 80.65 | 62.51 |
| Our Method | CSPDarknet53 | 3 | 83.01 | 70.65 | 50.49 | 27.14 |
| | | 4 | 90.12 | 88.36 | 81.27 | 34.38 |
| | | 5 | 88.85 | 88.15 | 85.34 | 68.13 |
| | | 6 | **90.29** | **90.13** | 86.43 | 71.74 |
| | | 7 | 89.98 | 89.26 | **87.42** | **74.68** |
| | | 8 | **90.50** | **89.68** | 86.32 | 71.40 |

*3) mAP:* Each category has a corresponding PRC, whose offline area is the average precision (AP), thus reflecting the model's global performance. The better the performance, the higher the precision corresponding to the same recall, i.e., a more prominent curve is located at the top right with a greater AP. The mAP is the mean of all categories' AP used to evaluate the model's detection performance against all categories. By setting the IoU thresholds of 0.5:0.05:0.95, we can calculate the mAP under various precision intervals. According to different calculation methods, mAP can be divided into mAP(07) and mAP(12). mAP(07) refers to using the VOC2007 evaluation metric, defined as follows:

$$\begin{cases} \text{mAP}(07) = \dfrac{1}{c}\displaystyle\sum_{i=1}^{c} \text{AP}_i(07) \\ \text{AP}(07) = \dfrac{1}{11}\displaystyle\sum_{r\in\{0,0.1,\ldots,1\}} p(r) \\ p(r) = \max\left[p\left(r'\right)\right],\ r' \geq r \end{cases} \tag{20}$$

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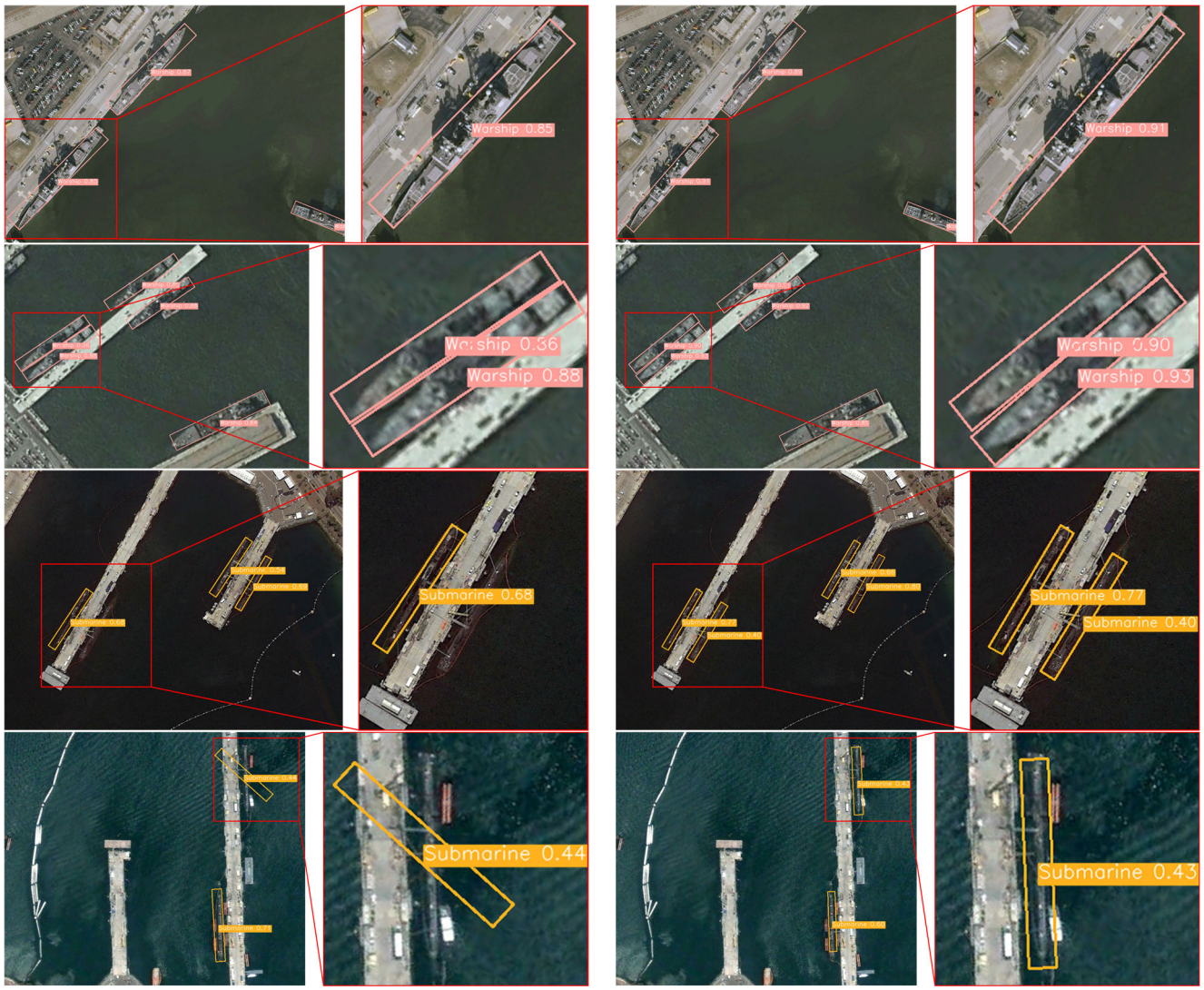



Fig. 10. Visualization of the detection results on the HRSC2016 test set: (a) detection results of BCL and (b) detection results of our method.

where $c$ is the category numbers, $p$ is the precision corresponding to recall $r$, and mAP(12) refers to using the VOC2012 evaluation metric. The definition is as follows:

$$\begin{cases} \text{mAP}(12) = \dfrac{1}{c}\displaystyle\sum_{i=1}^{c} \text{AP}_i(12) \\ \text{AP}(12) = \displaystyle\sum_{n=0} (r_{n+1} - r_n)\, p(r_{n+1}) \\ p(r_{n+1}) = \max\left[p\left(r'\right)\right], \ r' \geq r_{n+1} \end{cases} \tag{21}$$

where $n$ is the total number of samples. There is no essential difference between the two evaluation metrics except for the sampling density of PRC.

*4) Δ mAP:* In order to study the dominant precision interval of model positioning, mAP corresponding to IoU threshold of 0.5:0.05:0.95 was obtained by differential operation, which is defined as follows:

$$\Delta\text{mAP}_{\text{thr}} = \text{mAP}_{\text{thr}+0.05} - \text{mAP}_{\text{thr}} \tag{22}$$

where thr is the IoU threshold and $\Delta\text{mAP}_{\text{thr}}$ is the mAP corresponding to the IoU threshold range from thr to thr + 0.05. ΔmAP represents the accuracy distribution of different IoU threshold intervals, which can more intuitively measure the model's performance in each precision interval.

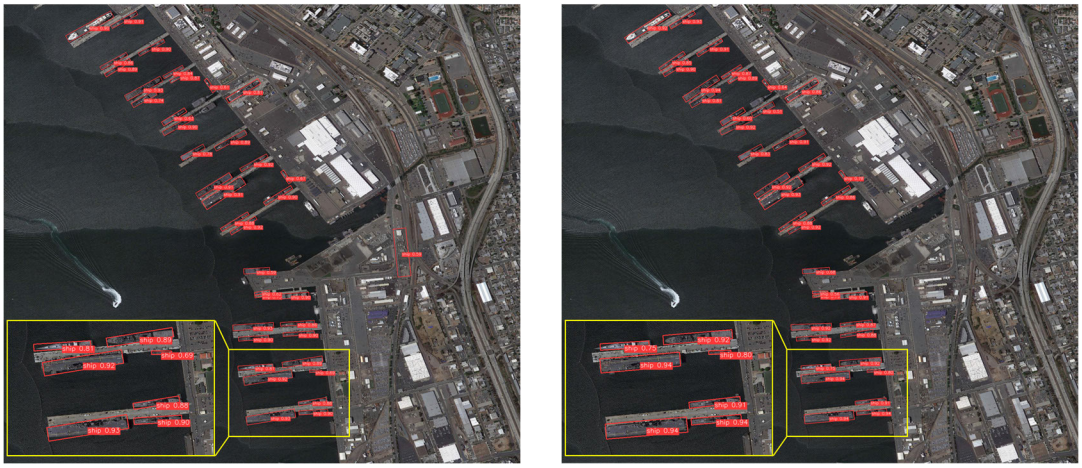
### D. Ablation Experiments

The ablation experiments are based on the control variable method and aim to verify our method, while BCL is the baseline model. RNMS is used to filter overlapping boxes.

*1) Influence of DAL:* BCL predicts the angles through dense coding, i.e., the longer the coding bits, the smaller the theoretical error. However, Yang et al. [46] claimed that the optimal effect is not positively correlated with code length. This is because the learning difficulty of the coding layers increases as the code length increases, affecting the overall learning effect.

We evaluate the DAL strategy on FGSD2021, the $L_1$ task of HRSC2016, and the DOTA-ship. The results reported in Table II show that the DAL strategy effectively improves the performance at different code lengths. Through latitude comparison, we find that DAL improves the detection

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(a) (b)

Fig. 11. Visualization of the detection results in a large image: (a) results of BCL and (b) results of our method.

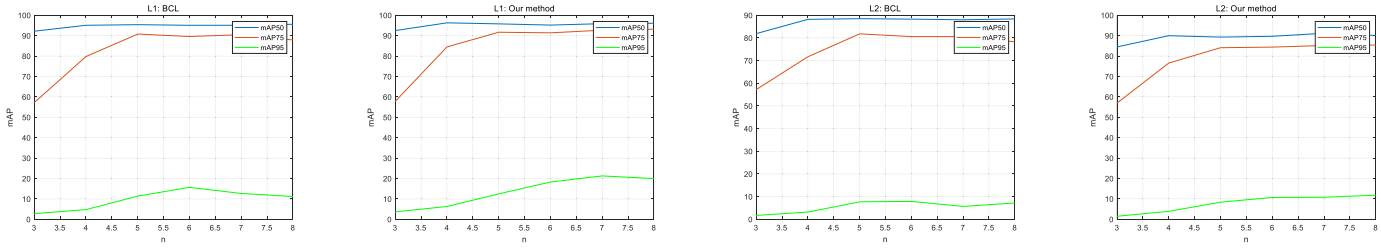



Fig. 12. mAp-$n$ curves. The curves represent $mAP_{50}$, $mAP_{75}$, and $mAP_{95}$ corresponding to different code lengths on $L_1$ and $L_2$ tasks.

performance of high-precision intervals and the learning effect of coding layers corresponding to small inter granularity. The mAP on the three datasets under different IoU thresholds is improved, and the increase in the high-precision interval is generally greater than that in the low-precision interval. We can see that $mAP_{95}$ on HRSC2016 increases the most by 7.9%/4.1%, and $mAP_{80}$ affords a maximum gain of 4.8%/3.9% and 3.3%/3.5% on FGSD2021 and DOTA-ship when $n = 8$, respectively, which are higher than the $mAP_{50}$ increase, proving DAL's applicability for high-precision detection. Additionally, the longitudinal comparison reveals that DAL significantly improves the performance of long-length code. When the code length becomes longer, the mAP's increased margin of each precision interval on the three datasets also shows an upward trend. In general, this experiment proves that guided learning can effectively reduce the long-length bits' learning difficulty and explore DAL's high-precision prediction potential, thus improving the overall performance.

*2) Influence of the Gradient Truncation Mechanism:* To demonstrate the effectiveness of the gradient truncation mechanism, we perform an ablation experiment on the $L_2$ task. First, we explore the influence of different truncation thresholds on detection performance, with the results reported in Table III. It can be seen that thr $= 0.7$ is a suitable choice. Specifically, we add the gradient truncation function to the angle loss as a control variable while the environmental and parameter settings are unchanged. The corresponding results are reported in Table IV, indicating that the model's overall performance is more advantageous when using a gradient truncation mechanism, and almost all precision intervals have a performance improvement. Therefore, we conclude that gradient truncation can balance the learning processes of samples with varying difficulty, effectively solve the overfitting problem, and is competent for bits prediction. It is worth noting that the gradient truncation mechanism is a universal method having no significant difference in the performance improvement of different length codes, while DAL is niche targeting and suitable for long-length bits performance mining.

### *E. Comparison With SOTA Methods*

*1) Results on FGSD2021:* FGSD2021 contains several ship targets with large aspect ratios and arbitrary orientation, which presents a significant challenge for the detection task. Table V highlights that our method achieves SOTA performance on FGSD2021, attaining the highest reported accuracy of 90.50%, which is 1.21% higher than center-head point detector (CHPDet). Table VI further shows the comparison of the detection results in the high-precision interval, inferring that the advantages of the proposed method are more significant in the high-precision interval. Indeed, on $mAP_{60}$, $mAP_{70}$, and $mAP_{80}$, our method achieves 1.15%, 0.85%, and 1.12% improvements compared with CHPDet, and 5.48%, 7.18%, and 6.74% improvements compared with rotation-equivariant detector (ReDet). Additionally, the detection performance of our method is also improved compared with the baseline under different code lengths, especially when the code length is considerable. Table VI highlights that when $n = 7$, our method reaches 74.68% on $mAP_{80}$, attaining a 7.63% gain compared with the baseline.

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*2) Results on HRSC2016:* We challenge the proposed method against SOTA methods on the $L_1$ and $L_2$ tasks. To evaluate our approach objectively, we still consider different code lengths. The corresponding results are reported in Tables VII and VIII. Specifically, Table VII demonstrates that our method's detection accuracy does not have a significant comparative advantage against the competitor methods when the IoU threshold is 0.5, at 89.9%/96.3% on $mAP_{50}$. As the IoU threshold increases, the advantage of our method becomes apparent. Compared with anchor-free corner evolution (ACE) [90], the performance of our method is 10.6% higher on $mAP_{75}$, and it has great advantages in the high-precision intervals compared with other methods. It is obvious that the proposed method achieves SOTA performance on both $mAP_{75}$ and $mAP_{95}$ (at 88.5%/93.2% and 13.5%/21.3%), and maintaining the lead on $mAP_{50:95}$ across all thresholds (at 75.6%/82.6%). The mAP of each code length shows varying degrees of improvement over the baseline network, especially in high-precision intervals. Indeed, the improvement tendency rises as the code length increases. Tables VII and VIII show that when $n$ reaches 7–8, the $mAP_{95}$ of the proposed method increases by 8%–9% and 4%–5%, and the $mAP_{50:95}$ increases by 4%–7% on both levels tasks, thus demonstrating our method's superiority in high-precision angle prediction.

### F. Visualization of the Proposed Method

This section visualizes the partial detection results of the baseline and the proposed methods on the HRSC2016 test set (Fig. 10). By comparison, it can be seen that our method predicts the ships' angles more precisely than BCL, especially for the submarine class that has an extreme aspect ratio. Additionally, we select a remote sensing image of Santiago port in Chile for large image detection, with an image size of $2653 \times 2353$ pixels. Before detection, we crop it into $800 \times 800$ subimages containing either an entire ship or a part, with an overlap of 400 pixels. The significant image detection result is obtained by merging the subimage detection results and filtering the redundant predicted bounding boxes through RNMS. The selected amplification region in Fig. 11 shows that the proposed method has a higher degree of coincidence with the target region and a smaller angle error, affording a superior positioning performance of large aspect ratio ship targets compared to BCL.

### G. Analysis and Discussion

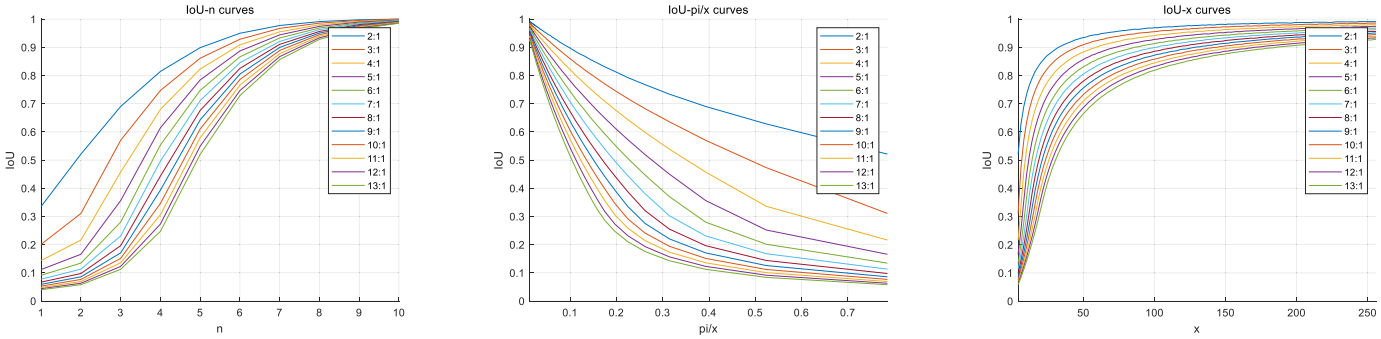
*1) Effect on Precision Distribution:* Tables VI–VIII report the detection results under various code lengths. Through longitudinal comparison, we find that neither our method's superiority is evident in the low-precision interval nor the improvement extent is significantly related to the code length. However, the proposed method's advantage is substantial in the high-precision intervals. As the code length increases, the mAP increases, with the optimal performance mainly concentrating on $n = 7$ and $n = 8$, the region where the maximum theoretical errors are $\pi/2^8$ and $\pi/2^9$, respectively. This compensates that the baseline method cannot exploit the potential of bits corresponding to small interval granularity.

When $n$ is in the 3–5 range, both the baseline and proposed methods have a mAP spike in the high-precision interval, as shown in Fig. 12. This is because when $n$ is in the 3–5 range, a short code length and a large angle granularity interval exist. In this case, increasing $n$ significantly improves the theoretical precision. To validate this conjecture, we simulate the relationship between the IoU of two rotated boxes and three variables under different aspect ratios, as shown in Fig. 13. The experimental conditions assume that the rotated boxes share the same center, size, and aspect ratios. From the IoU-$n$ curves in Fig. 13(a), we can see that with the increase of code length, IoU gradually increases, and the growth rate gets the highest value when $n$ is in the range of 3–5. We set the angle deviation $\Delta\theta = \pi/x$ and obtain the IoU $- \pi/x$ curves and IoU-$x$ curves, as illustrated in Fig. 13(b) and (c), respectively. Fig. 13(b) shows that there is a negative correlation between IoU and angle deviation, and the reduction rate of IoU near zero deviation is related to the aspect ratio. The larger the aspect ratio is, the faster the IoU decreases near the zero deviation, that is, the more sensitive it is to the angle change. Compared with Fig. 13(b), Fig. 13(c) can more clearly observe the changes of IoU in the high-precision intervals. The code lengths in 3–5 correspond to the $x$ interval 0–50. From Fig. 13(c), it is evident that the curve has a significant slope in this interval, which means that the IoU is sensitive to code length changes. Indeed, as the code length increases, the theoretical precision's improvement speed gradually slows, and the change of the high-precision interval detection results stabilizes, thus proving the above conjecture.

To further explore our method's effectiveness on the mAP interval, in Fig. 14, we visualize the mAP distribution and probability density distribution of $L_1$ and $L_2$ tasks on HRSC2016. The mAP distribution curve reveals that the proposed method's mAP is higher than that of the baseline method under different code lengths. We obtain the probability density distribution through the forward difference and negation operation of the cumulative distribution, i.e., the mAP distribution of each IoU interval. Our method presents a more significant advantage over BCL in the high-precision intervals, especially for the probability density distribution in the 0.90–0.95 and 0.95–1 intervals. This indicates that the IoU distribution of the predicted and ground-truth boxes is more concentrated in the high-precision intervals, proving DAL's guiding effect on the learning process of the coding layers that correspond to the small interval granularity. Fig. 14 reveals that the peak value of the probability density distribution is mainly in the range of 0.90–0.95. In order to further centralize the distribution to the range of 0.95–1, it is essential to improve the learning effect of angle coding layers except for the coding bits' theoretical precision and the model's performance.

By comparing the probability density distribution of the four classes on the $L_2$ task of HRSC2016, we find that the submarine's mAP distribution is relatively scattered. This is because the submarine has an extreme aspect ratio and demands the positioning precision of a rotating bounding box. In addition, the submarine-background color contrast is low, and these factors further increase its detection difficulty

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(a) (b) (c)

Fig. 13. Curves represent the IoU values corresponding to different variables between two arbitrary-OBBs of the same center and size. Different colors indicate different aspect ratios: (a) relationship between IoU and code length; (b) relationship between IoU and angle deviation; and (c) relationship between IoU and $x$.

TABLE VII
DETECTION ACCURACY ON THE $L_1$ DETECTION TASK OF HRSC2016

| Method | Backbone | $n$ | $mAP_{50}$(07/12) | $mAP_{75}$(07/12) | $mAP_{95}$(07/12) | $mAP_{50:95}$(07/12) |
|---|---|---|---|---|---|---|
| $R^2$CNN [60] | ResNet101 | - | 73.1/79.7 | - | - | - |
| RRPN [38] | ResNet101 | - | 79.1/85.6 | - | - | - |
| RRD [78] | VGG16 | - | 84.3/- | - | - | - |
| DRN [61] | Hourglass104 | - | 84.3/92.7 | - | - | - |
| RetinaNet-KFIoU [79] | ResNet50 | - | 84.4/- | 58.3/- | - | 51.3/- |
| RetinaNet-GWD [80] | ResNet50 | - | 85.6/- | 60.3/- | - | 52.9/- |
| RetinaNet-KLD [81] | ResNet50 | - | 87.5/- | 72.4/- | - | 57.8/- |
| GRS-Det [82] | ResNet50 | - | 88.9/- | - | - | - |
| CPS-Det [83] | ResNet50 | - | 89.1/- | - | - | - |
| CenterMap-Net [84] | ResNet50 | - | -/92.8 | - | - | - |
| RetinaNet-R [73] | ResNet101 | - | 89.2/95.2 | - | - | - |
| $R^3$Det [73] | ResNet101 | - | 89.3/96.0 | - | - | - |
| $R^3$Det-GWD [80] | ResNet50 | - | 89.4/- | 65.9/- | - | 56.1/- |
| FPN-CSL [45] | ResNet101 | - | 89.6/96.1 | - | - | - |
| OrtDet [85] | - | - | 89.6/- | - | - | - |
| RetinaNet-DSLA [86] | ResNet101 | - | 89.6/- | - | - | - |
| FR-Est [62] | ResNet101 | - | 89.7/- | - | - | - |
| Yolov5m-CSL [45] | CSPDarknet53 | 180 | 89.7/96.2 | 87.8/92.0 | 8.5/13.2 | 72.9/79.9 |
| $TS^4$Net [87] | ResNet101 | - | 89.8/94.7 | - | - | - |
| $R^3$Det-KLD [81] | ResNet50 | - | 90.0/- | 77.4/- | - | 61.4/- |
| OSKDet [88] | ResNet101 | - | 90.0/- | - | - | - |
| OA-Det [89] | ResNet101 | - | **90.1**/- | - | - | - |
| ACE [90] | DLA34 | - | **90.5**/**96.7** | 77.9/- | - | - |
| Baseline | CSPDarknet53 | 3 | 87.9/92.2 | 47.5/57.2 | 3.0/2.8 | 48.4/55.3 |
| | | 4 | 89.5/95.1 | 73.3/79.8 | 2.3/4.8 | 63.8/70.0 |
| | | 5 | 89.6/95.4 | 79.7/90.8 | 6.2/11.4 | 71.2/78.3 |
| | | 6 | 89.4/95.1 | 78.8/89.6 | 9.1/15.7 | 71.5/78.9 |
| | | 7 | 89.5/95.1 | 79.2/90.5 | 6.2/12.7 | 71.0/78.2 |
| | | 8 | 89.5/95.6 | 77.7/87.9 | 6.7/11.2 | 69.4/76.0 |
| Our Method | CSPDarknet53 | 3 | 87.8/92.5 | 47.5/57.8 | 2.3/3.7 | 48.0/55.4 |
| | | 4 | 89.8/**96.3** | 76.0/84.5 | 10.0/6.3 | 65.5/72.3 |
| | | 5 | 89.7/95.8 | 87.5/91.7 | 6.9/12.5 | 72.2/79.1 |
| | | 6 | 89.4/95.2 | 87.3/91.4 | 11.1/18.3 | **75.0**/81.5 |
| | | 7 | 89.7/95.9 | **88.1**/**92.7** | **13.5**/**21.3** | **75.6**/**82.5** |
| | | 8 | 89.9/96.1 | **88.5**/**93.2** | **12.6**/**20.0** | **75.6**/**82.6** |

compared to the other three classes. To further analyze this case, we argue from the perspective of the PRC. As illustrated in Fig. 15, the precision and recall of the submarine class decline sharply with the increase of the IoU threshold, while the declining tendency of the other classes is much more moderate. From the reverse change of the IoU threshold, we find that the PRCs of the four classes experience a rapid upward phase and then stabilize. The difference lies in the rising range between 0.80 and 0.85 for the submarine and 0.85–0.90 for the remaining classes. This confirms the probability density distribution depicted in Fig. 14.

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TABLE VIII
DETECTION ACCURACY ON THE $L_2$ DETECTION TASK OF HRSC2016

| Method | Backbone | $n$ | Air. | War. | Mer. | Sub. | $mAP_{50}$ (07/12) | $mAP_{75}$ (07/12) | $mAP_{95}$ (07/12) | $mAP_{50:95}$ (07/12) |
|---|---|---|---|---|---|---|---|---|---|---|
| BBAVectors [74] | ResNet101 | - | 82.5/86.0 | 87.9/93.9 | 68.8/74.7 | 65.8/75.7 | 76.3/82.6 | 39.7/46.9 | 0.2/0.1 | 41.6/46.5 |
| Yolov5m-CSL [45] | CSPDarknet53 | 180 | 78.9/91.0 | **88.5**/95.7 | 73.8/81.2 | 80.5/89.0 | 80.4/89.2 | 60.6/70.2 | 3.1/4.8 | 55.5/63.8 |
| Baseline | CSPDarknet53 | 3 | **88.1**/92.4 | 84.5/89.0 | 70.6/77.9 | 54.5/68.2 | 74.4/81.9 | 46.5/57.2 | 1.6/1.7 | 42.6/50.4 |
| | | 4 | 78.2/90.2 | 87.1/93.6 | 69.6/76.5 | 81.8/92.2 | 79.2/88.2 | 61.1/71.7 | 1.2/3.2 | 54.0/62.7 |
| | | 5 | 79.2/90.3 | 87.7/94.5 | 70.9/78.4 | 80.9/91.0 | 79.7/88.5 | 74.0/81.8 | 4.9/7.7 | 61.9/70.6 |
| | | 6 | 87.9/**93.1** | 87.0/93.6 | 72.3/79.9 | 72.7/86.3 | 80.0/88.3 | 72.4/80.6 | 5.5/7.9 | 60.8/68.9 |
| | | 7 | 86.2/90.4 | 87.5/94.4 | 70.3/78.1 | 80.7/89.2 | 81.2/88.0 | 71.9/80.6 | 5.3/5.7 | 61.2/68.8 |
| | | 8 | **88.1**/92.5 | 87.5/94.5 | 72.4/80.0 | 71.4/86.5 | 79.8/88.4 | 69.2/78.3 | 7.3/7.2 | 59.9/67.7 |
| Our Method | CSPDarknet53 | 3 | **88.4**/**92.7** | 85.5/90.1 | 72.4/79.9 | 64.9/75.2 | 77.8/84.5 | 46.6/57.0 | 2.7/1.5 | 45.2/52.3 |
| | | 4 | 86.5/91.1 | 87.3/94.1 | **73.9**/82.1 | 81.8/92.8 | 82.4/90.0 | 65.8/76.6 | 2.2/3.9 | 58.3/65.6 |
| | | 5 | 87.2/91.5 | **88.0**/**95.0** | 73.3/80.8 | 81.3/89.8 | 82.4/89.3 | **77.6**/84.1 | 4.2/8.4 | 64.3/71.4 |
| | | 6 | 87.5/91.9 | 87.1/93.6 | **74.5**/**82.7** | 81.6/90.5 | 82.7/89.7 | 75.7/84.4 | **9.2**/10.7 | 65.1/72.6 |
| | | 7 | 87.1/**93.1** | 87.6/94.6 | 73.7/**82.6** | **90.4**/**94.5** | **84.7**/**91.2** | **77.7**/**85.2** | 5.8/**10.8** | **65.9**/**73.6** |
| | | 8 | 87.1/92.4 | 87.4/94.3 | 71.0/79.1 | **90.6**/**94.7** | **84.0**/**90.1** | **77.7**/**85.4** | **9.8**/**11.8** | **66.2**/**73.4** |

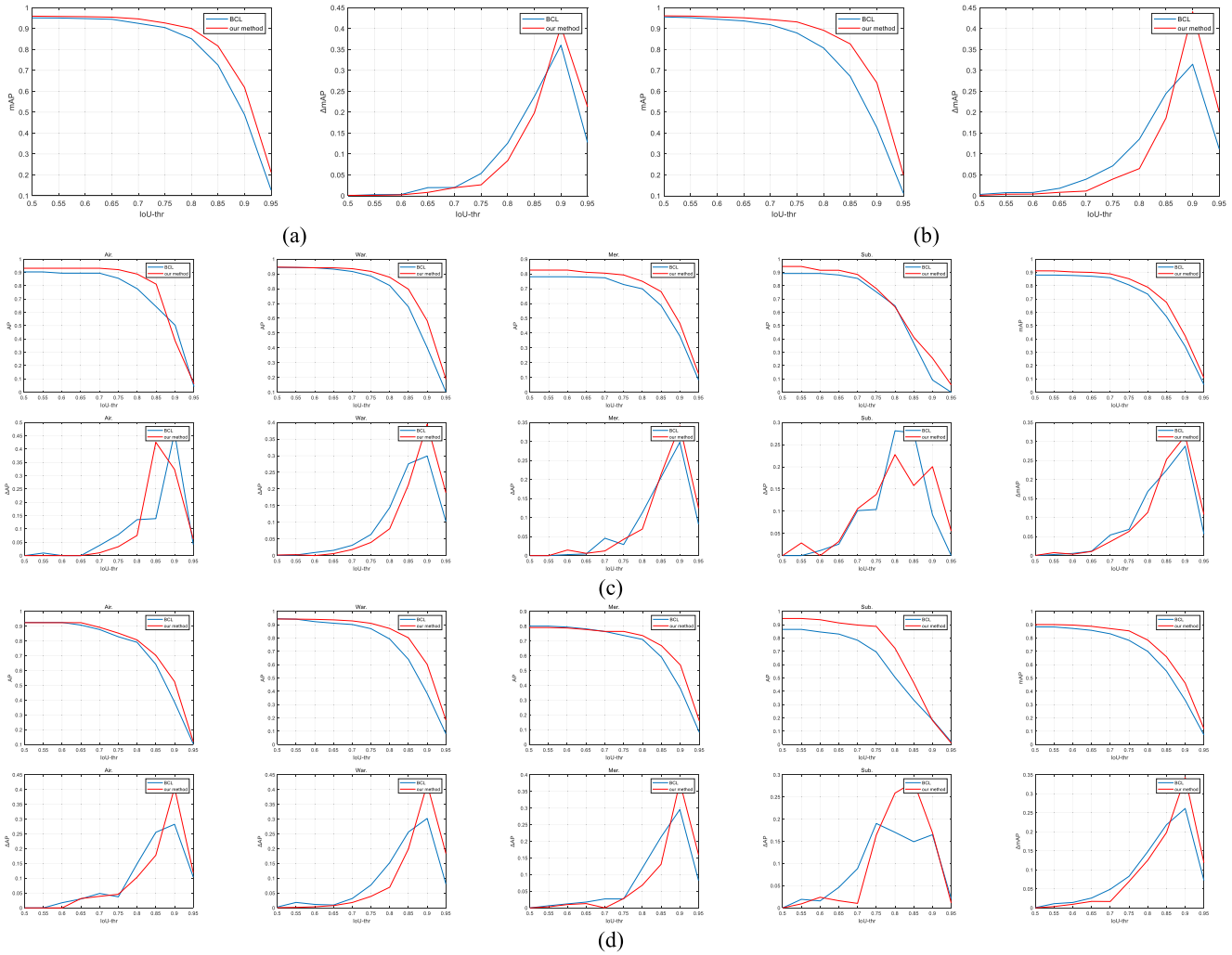



Fig. 14. Cumulative distribution and probability density distribution of mAP on $L_1$ and $L_2$ tasks: (a) $L_1$: $n = 7$; (b) $L_1$: $n = 8$; (c) $L_2$: $n = 7$; and (d) $L_2$: $n = 8$.

*2) Effect on Angle Learning:* This section explores the angle learning effect. Specifically, we statistically evaluate the convergence of the coding bit error rate and the average angle error during the training process (Fig. 16). The average angle error is defined as follows:

$$\Delta\theta = \frac{1}{N_{\mathrm{GT}}} \sum_{i}^{N_{\mathrm{GT}}} \Delta\theta_i \tag{23}$$

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TABLE IX
CODING BIT ERROR RATE, AVERAGE ANGLE ERROR, AND ANGLE CONTRIBUTION RATE ON THE HRSC2016 AND FGSD2021 TEST SETS

| Dataset | Method | n | b1 | b2 | b3 | b4 | b5 | b6 | b7 | b8 | $\Delta\theta$ | $C_\theta$ |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| HRSC2016 | Baseline | 3 | 0.0068 | 0.0122 | 0.0353 | - | - | - | - | - | 6.21 | 58.09 |
| | | 4 | 0.0071 | 0.0122 | 0.0403 | 0.0744 | - | - | - | - | 3.18 | 105.30 |
| | | 5 | 0.0074 | 0.0146 | 0.0391 | 0.0745 | 0.1582 | - | - | - | 2.10 | 184.94 |
| | | 6 | **0.0059** | **0.0119** | **0.0349** | **0.0727** | **0.1549** | 0.4344 | - | - | 2.25 | 167.39 |
| | | 7 | 0.0070 | 0.0125 | 0.0396 | 0.0789 | 0.1619 | 0.4335 | 0.4786 | - | 2.42 | 168.27 |
| | | 8 | 0.0081 | 0.0147 | 0.0372 | 0.0736 | 0.1604 | 0.4729 | **0.4780** | 0.5252 | 2.73 | 159.66 |
| | Our Method | 3 | 0.0071 | 0.0132 | 0.0378 | - | - | - | - | - | 6.12 | 58.11 |
| | | 4 | 0.0076 | 0.0156 | 0.0388 | 0.0775 | - | - | - | - | 3.05 | 105.15 |
| | | 5 | 0.0076 | 0.0147 | 0.0418 | 0.0778 | 0.1744 | - | - | - | 1.89 | 182.38 |
| | | 6 | 0.0109 | 0.0210 | 0.0452 | 0.0867 | 0.1801 | 0.3964 | - | - | 1.86 | 194.09 |
| | | 7 | 0.0084 | 0.0138 | 0.0394 | 0.0759 | 0.1737 | **0.3654** | 0.4961 | - | **1.71** | **208.55** |
| | | 8 | 0.0096 | 0.0173 | 0.0440 | 0.0832 | 0.1730 | 0.3828 | 0.5092 | **0.5036** | 1.77 | 205.09 |
| FGSD2021 | Baseline | 3 | **0.0081** | 0.0094 | **0.0148** | - | - | - | - | - | 4.52 | 96.68 |
| | | 4 | 0.0255 | 0.0280 | 0.0355 | **0.0741** | - | - | - | - | 3.14 | 113.90 |
| | | 5 | 0.0313 | 0.0369 | 0.0457 | 0.0907 | 0.1489 | - | - | - | 1.68 | 241.07 |
| | | 6 | 0.0287 | 0.0331 | 0.0393 | 0.0855 | 0.1379 | 0.3082 | - | - | 1.33 | 318.88 |
| | | 7 | 0.0262 | 0.0312 | 0.0394 | 0.0818 | 0.1387 | 0.2986 | **0.4560** | - | 1.57 | 302.03 |
| | | 8 | 0.0383 | 0.0364 | 0.0427 | 0.0834 | 0.1499 | 0.3381 | 0.4868 | **0.4875** | 1.88 | 251.50 |
| | Our Method | 3 | 0.0088 | **0.0088** | 0.0189 | - | - | - | - | - | 4.56 | 97.49 |
| | | 4 | 0.0179 | 0.0234 | 0.0339 | 0.0819 | - | - | - | - | 3.20 | 99.35 |
| | | 5 | 0.0279 | 0.0316 | 0.0403 | 0.0856 | 0.1365 | - | - | - | 1.62 | 235.46 |
| | | 6 | 0.0303 | 0..0371 | 0.0439 | 0.0896 | 0.1391 | 0.2645 | - | - | 1.11 | 358.42 |
| | | 7 | 0.0330 | 0.0367 | 0.0454 | 0.0821 | **0.1350** | **0.2519** | 0.4658 | - | **1.08** | **368.30** |
| | | 8 | 0.0217 | 0.0267 | 0.0384 | 0.0806 | 0.1420 | 0.261 | 0.4786 | 0.4885 | 1.15 | 359.22 |

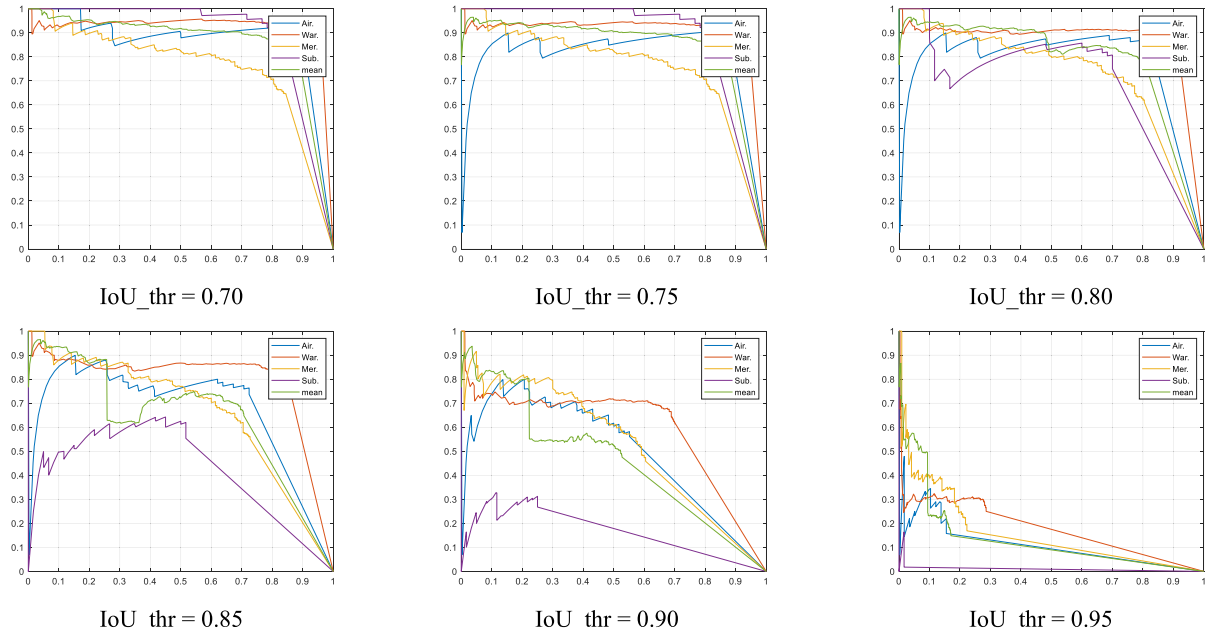



Fig. 15. PRCs at different IoU thresholds on $L_2$ task. Four colors indicate aircraft carrier, warship, merchant ship, and submarine, respectively.

where $N_{GT}$ is the number of ground-truth labels. $\Delta\theta_i$ indicates the long edge acute angle between the predicted and ground-truth boxes, and its upper limit is 90°.

Fig. 16 highlights that a higher bit has a faster convergence rate and a better learning effect, conforming to the above analysis on the angle interval granularity. Compared with BCL, our method's coding bit error rate is slightly higher, and the average angle error is smaller. This is because we focus on the relevance and integrity of coding, and the low bits are adjusted according to the prediction of high bits rather than focusing on the absolute consistency with the ground-truth labels. To better demonstrate the advantages of our method in angle prediction, we define the angle contribution rate to measure the distribution of small error intervals

$$C_\theta = \frac{1}{N_{GT}} \sum_{i}^{N_{GT}} \left( \frac{90}{\Delta\theta_i} - 1 \right). \tag{24}$$

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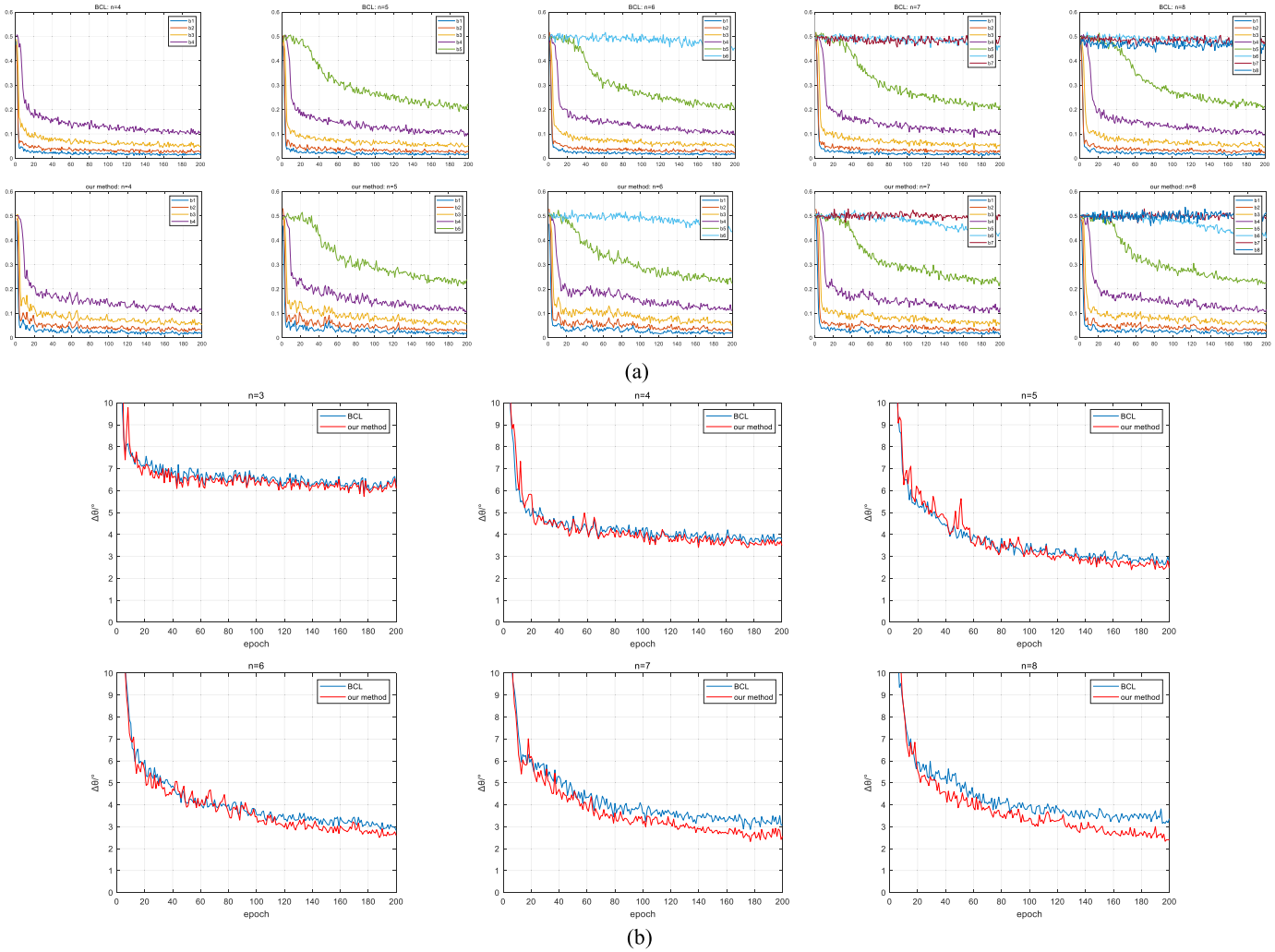



Fig. 16. Convergence curves of: (a) coding bit error rate and (b) average angle error in the training process on HRSC2016. In (a) and (b), the first-row curves are generated by BCL, and the second-row curves are generated by our method.

We compare the angle prediction performance of the baseline and proposed methods on a test set of HRSC2016. The experimental results at the top half of Table IX show that comparing the coding bit error rate is consistent with the convergence curve in the training process. The average angle error and the angle contribution rate of the baseline are optimal when the code length is 5. In contrast, our angle localization performance improves with the increase of code length, which is mutually verified with the angle convergence curve in Fig. 16(b) and further proves the guided learning superiority of the DAL strategy. In addition, to verify our method's generalization ability, we conducted experiments on FGSD2021, with the corresponding results reported at the bottom half of Table IX. It can be seen that when the code length is 7 and 8, the average angle error reduces from 1.57 and 1.88 to 1.08 and 1.15, and the angle contribution rate of the proposed method increases from 302.03 and 251.50 to 368.30 and 359.22, respectively. The variation trend of the increase with the code length is the same as in HRSC2016. This proves our guided learning strategy's generalization ability.

## V. Conclusion

This article proposes a rotation-based ship detection algorithm based on angle coding. The proposed method affords appealing adaptability to large aspect ratio ships and effectively deals with arbitrary orientation and dense arrangement problems. To solve the problem of angle coding bits with weak correlation in BCL, we propose the DAL strategy, which dynamically adjusts the subsequent coding bits labels according to the prediction results of the high coding bits and guides the code to perform tendency learning. Such a training strategy breaks the conventional training mode strictly limited to ground-truth labels, improves the overall coupling of coding bits, and can be better qualified for detecting rotated objects. Furthermore, we add a gradient truncation function to the angle loss to prevent the over-fitting problem of the high coding bits caused by the difference in the learning difficulty. This strategy improves the learning effect of the low coding bits and enhances the perception ability of angle codes to small granularity intervals. The experimental results demonstrate that the proposed method can effectively improve the precision of angle prediction and is suitable for detecting

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large aspect ratio targets like ships. Future work will further explore the differences in learning difficulty between coding bits and propose a more effective balancing mechanism to improve the problem of low coding bits' and poor learning.

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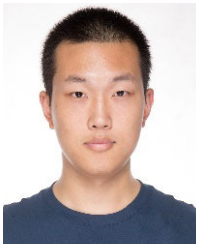
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